

XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

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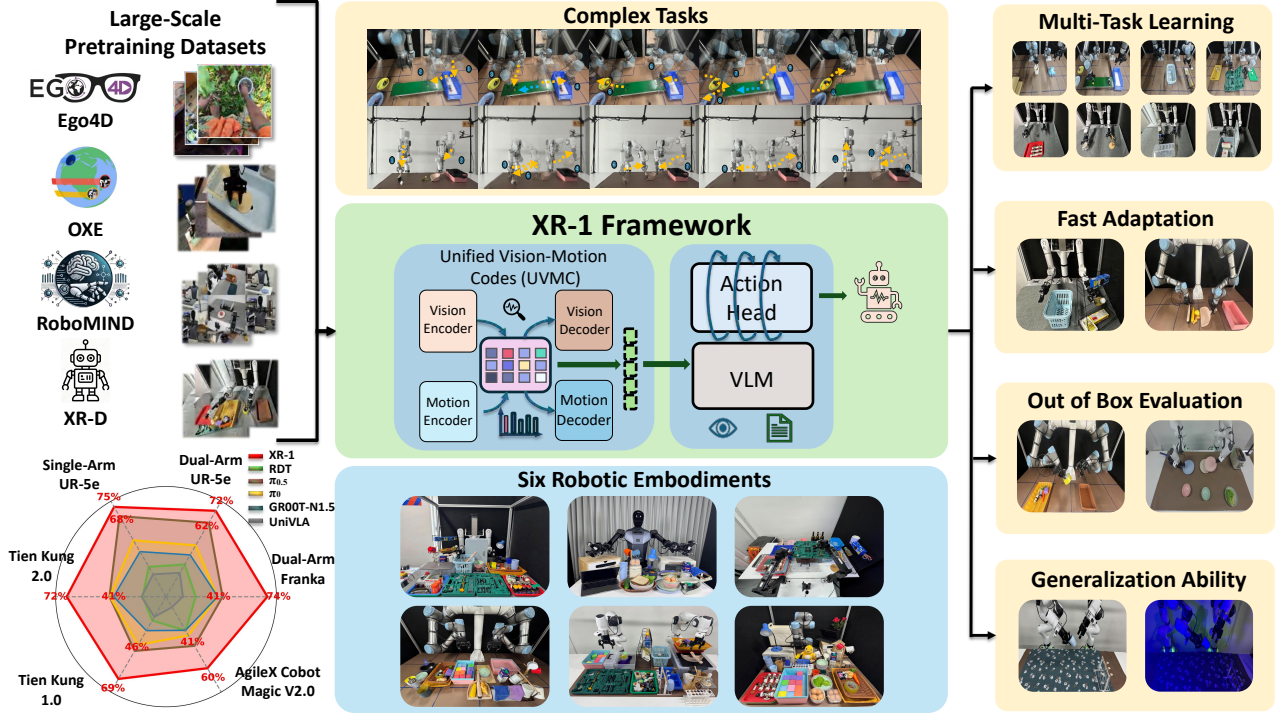


Figure 1. We introduce **X Robotic Model 1 (XR-1)**, a versatile and scalable vision-language-action framework. XR-1 supports robust multi-task learning across diverse robot embodiments and environments.

Abstract

Recent progress in large-scale robotic datasets and vision-language models (VLMs) has advanced research on vision-language-action (VLA) models. However, existing VLA models still face two fundamental challenges: (i) producing precise low-level actions from high-dimensional ob-

servations, (ii) bridging domain gaps across heterogeneous data sources, including diverse robot embodiments and human demonstrations. Existing methods often encode latent variables from either visual dynamics or robotic actions to guide policy learning, but they fail to fully exploit the complementary multi-modal knowledge present in large-scale, heterogeneous datasets. In this work, we present **X Robotic Model 1 (XR-1)**, a novel framework for versatile and scalable VLA learning across diverse robots, tasks, and environments. At its core, XR-1 introduces the *Unified Vision-Motion Codes (UVMC)*, a discrete latent representation learned via a dual-branch VQ-VAE that jointly encodes visual dynamics and robotic motion. UVMC addresses these challenges by (i) serving as an intermediate representation between the observations and actions, and (ii) aligning mul-

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timodal dynamic information from heterogeneous data sources to capture complementary knowledge. To effectively exploit UVMC, we propose a *three-stage training paradigm*: (i) self-supervised UVMC learning, (ii) UVMC-guided pretraining on large-scale cross-embodiment robotic datasets, and (iii) task-specific post-training. We validate XR-1 through extensive real-world experiments with more than 14,000 rollouts on six different robot embodiments, spanning over 120 diverse manipulation tasks. XR-1 consistently outperforms state-of-the-art baselines such as $\pi_{0.5}$, π_0 , RDT, UniVLA, and GR00T-N1.5 while demonstrating strong generalization to novel objects, background variations, distractors, and illumination changes. Our project is at <https://xr-1-vla.github.io/>.

1. Introduction

The long-term goal of Embodied AI (Pfeifer & Iida, 2004) is to develop generalist agents capable of executing diverse natural language instructions across real-world settings, such as households, factories, and hospitals. Recent advancements in Vision-Language Models (VLMs) (Bai et al., 2023; Gao et al., 2024; Li et al., 2023; Liu et al., 2023; Zhang et al., 2024; Beyer et al., 2024; Wang et al., 2024b) demonstrate that large-scale pre-training on internet data provides robust visual-semantic understanding. Building on this, Vision-Language-Action (VLA) models (Zitkovich et al., 2023; Kim et al., 2024; Black et al., 2025; Liu et al., 2025; Wen et al., 2025a; Cheang et al., 2025a; Liu et al., 2026a; Lee et al., 2025; Intelligence et al., 2025; Bu et al., 2025b) incorporate action heads to ground perception and language into executable motor commands.

VLA training typically follows a two-stage pipeline: (i) large-scale pre-training on cross-embodiment datasets (Walke et al., 2023; O’Neill et al., 2024; Wu et al., 2025a) to learn general priors, and (ii) task-specific post-training. Despite VLM advancements, two challenges persist: (i) Precision Gap: Mapping high-dimensional observations to precise low-level actions is difficult due to multimodal uncertainty; even centimeter-level errors lead to failure in dexterous or contact-rich tasks. (ii) Data Heterogeneity: Cross-embodiment transfer is hindered by morphological differences (hardware and DoF) and the lack of explicit action labels or visual alignment in human demonstration videos.

To address these challenges, prior research (Cui et al., 2023; Shafiullah et al., 2022; Lee et al., 2024; Xie et al., 2025; Zheng et al., 2025) has explored latent representations as intermediate abstractions. Current directions either encode ac-

tion sequences for motion modeling (Shafiullah et al., 2022; Wu et al., 2025b; Bauer et al., 2025), which requires costly labeled data, or encode visual dynamics from videos (Cui et al., 2023; Hu et al., 2025; Bu et al., 2025a), which lacks explicit action grounding. Both approaches treat vision and action in isolation, failing to capture the multimodal alignment necessary for task-relevant correspondences. In contrast, humans fuse heterogeneous sensory inputs into *supramodal codes* (Park et al., 2025), abstracting embodiment-specific details while preserving task semantics. Inspired by this, we argue that robotic representation learning must move beyond unimodal abstractions toward a joint encoding of visual dynamics and motor control.

Addressing the limitations of unimodal representations and inspired by human supramodal cognition, we propose X Robotic Model 1 (XR-1) to achieve cross-data exploitation and cross-embodiment control. Central to our framework is Unified Vision-Motion Codes (UVMC), a discrete latent representation that jointly captures visual dynamics and robotic motion. UVMC is learned via a dual-branch VQ-VAE where vision and motion branches share a unified latent space to enforce cross-modal consistency. To suppress task-irrelevant visual noise and ensure the extraction of motion-centric features, we introduce a vision-motion alignment loss that regularizes visual codes to align with their corresponding motion counterparts. Building upon UVMC, XR-1 employs a three-stage paradigm: (i) self-supervised UVMC learning on large-scale robot and human video datasets; (ii) cross-embodiment pre-training that injects UVMC knowledge into a VLM via learnable tokens; and (iii) task-specific post-training for refinement. This architecture allows XR-1 to leverage heterogeneous data while ensuring embodiment-agnostic consistency.

We extensively evaluate XR-1 through more than 14k rollouts across six distinct robot embodiments, including Tien Kung 1.0/2.0, Single-/Dual-Arm UR-5e, Dual-Arm Franka, and AgileX Cobot Magic 2.0, covering over 120 manipulation tasks. XR-1 outperforms state-of-the-art baselines such as $\pi_{0.5}$, π_0 , RDT, UniVLA, and GR00T-N1.5 across challenging scenarios involving bimanual collaboration, dexterous manipulation, deformable objects, contact-rich interactions, dynamic settings, and long-horizon manipulation. Our main contributions are summarized as follows:

- We propose **X Robotic Model 1 (XR-1)**, a scalable three-stage framework for VLA learning that effectively leverages heterogeneous data sources, including Internet-scale human videos and diverse robot datasets, and integrates with diverse VLA architectures.
- We introduce the *Unified Vision-Motion Codes (UVMC)*, a discrete latent representation that encodes both environmental dynamics and robotic motion,

while an alignment loss enforces consistent multimodal embeddings across embodiments via UVMC.

- We validate XR-1 with over 14,000 real-world rollouts on six robot embodiments across 123 tasks, and demonstrate that it consistently outperforms strong baselines such as $\pi_{0.5}$, π_0 , RDT, UniVLA, and GR00T-N1.5.

2. Related Work

2.1. Vision-Language-Action Models

Modern robotics aims to develop robust, general-purpose Vision-Language-Action (VLA) policies capable of zero-shot cross-embodiment transfer. While early Imitation Learning (IL) methods relied on narrow demonstrations (Cui et al., 2023; Zhao et al., 2023; Chi et al., 2023; Ze et al., 2024; Fu et al., 2024; Bharadhwaj et al., 2024; Ze et al., 2024; Cao et al., 2025; Su et al., 2025; Fan et al., 2025), scalability and generalization remained limited. This bottleneck has been addressed by large-scale datasets, such as BridgeData (Ebert et al., 2022; Walke et al., 2023), DROID (Khazatsky et al., 2024), Open X-Embodiment (O’Neill et al., 2024), RoboMIND (Wu et al., 2025a), and AgiBot World (Bu et al., 2025a), which catalyzed the transition to generalist policies. Key advancements include RT-1 (Brohan et al., 2023) and RT-2 (Zitkovich et al., 2023), which unified vision-language pre-training with action generation. Subsequent frameworks like RT-X (O’Neill et al., 2024) and Octo (Team et al., 2024b) further embraced data heterogeneity, while PaLM-E (Driess et al., 2023) utilized visual-conditioned LLMs to enhance complex task planning and semantic grounding.

Beyond core training paradigms, research has focused on augmenting VLA models with extensive world knowledge and diverse capabilities through internet-scale representations. Recent advancements include CrossFormer (Zhang & Yan, 2023), OpenVLA (Kim et al., 2024), HPT (Wang et al., 2024a), π_0 (Black et al., 2025), RDT (Liu et al., 2025), TinyVLA (Wen et al., 2025b), GR00T (Bjorck et al., 2025), HybridVLA (Liu et al., 2026a), SwitchVLA (Li et al., 2025a), MLA (Liu et al., 2026b), and X-VLA (Zheng et al., 2026). Specifically, $\pi_{0.5}$ (Intelligence et al., 2025) and FSD (Yuan et al., 2026) leverage image-text pre-training for grounding, while CoT-VLA (Zhao et al., 2025) and InstructVLA (Yang et al., 2026) enhance reasoning and instruction following. Additionally, SpatialVLA (Qu et al., 2025) improves spatial awareness, and Diffusion-VLA (Wen et al., 2025c) utilizes generative modeling for action generation. Our framework, XR-1, introduces a novel three-stage process that synergizes human and robot data, departing from traditional two-stage paradigms. Its core is an initial self-supervised stage that learns unified vision-motion representations, which serve as auxiliary features to optimize sub-

sequent large-scale VLA pre-training. This model-agnostic approach ensures flexibility, which we validate by building upon base models like π_0 and SwitchVLA to develop the high-performing XR-1 and efficient XR-1-Light.

2.2. Latent Representation Learning

A primary bottleneck in learning robust visuomotor policies is the gap between high-dimensional pixel observations and low-dimensional motor commands. To bridge this, prior research (Cui et al., 2023; Shafiullah et al., 2022; Lee et al., 2024; Zheng et al., 2025; Xie et al., 2025) utilizes latent representations as a critical intermediary layer. Current methodologies generally fall into two unimodal categories, with the first focusing on modeling low-level motor dynamics by discretizing continuous actions into latent tokens. This tokenization approach, pioneered by Behavior Transformers (BeT) (Shafiullah et al., 2022) and refined by QueST (Mete et al., 2024), transforms continuous control into a sequence-generation task. Recent advancements have further optimized this direction: (Bauer et al., 2025) introduced discrete latent actions for data efficiency, while ATE (Zhang et al., 2026) improved alignment between visual inputs and discretized action spaces. Similarly, Moto (Chen et al., 2025) and Discrete Policy (Wu et al., 2025b) demonstrate that action tokenization scales effectively for generalized control. However, these action-centric methods are fundamentally limited by dependence on vast quantities of labeled robotic data, which is prohibitively expensive to acquire at scale. Alternatively, the second category exploits unlabeled video data (Cui et al., 2023; Du et al., 2023; Hu et al., 2025; He et al., 2024; Ye et al., 2025; Cheang et al., 2025b) to encode visual flow and state transitions. While models like C-BeT (Cui et al., 2023) and UniPi (Du et al., 2023) utilize "play" data or video generation, more recent works—including VPP (Hu et al., 2025), LAPA (Ye et al., 2025), GR-2 (Cheang et al., 2025b), VPDD (He et al., 2024), and GO-1 (Bu et al., 2025a)—pre-train on actionless human and web videos to capture task semantics. However, these approaches, alongside UVA (Li et al., 2025b), often isolate vision from action during representation learning. This reliance on action-free data lacks explicit grounding, creating a critical alignment gap between understanding visual changes and executing the fine-grained motor control required to achieve them.

Crucially, by treating vision and action in isolation, existing unimodal paradigms miss the causal link between observation and execution. We address this limitation with XR-1, which introduces Unified Vision-Motion Codes (UVMC), a discrete latent representation learned jointly from visual dynamics and robotic motion. By constructing a bimodal latent space, UVMC encodes the cause-and-effect relationship between seeing and acting, abstracting away embodiment-specific details while preserving task semantics.

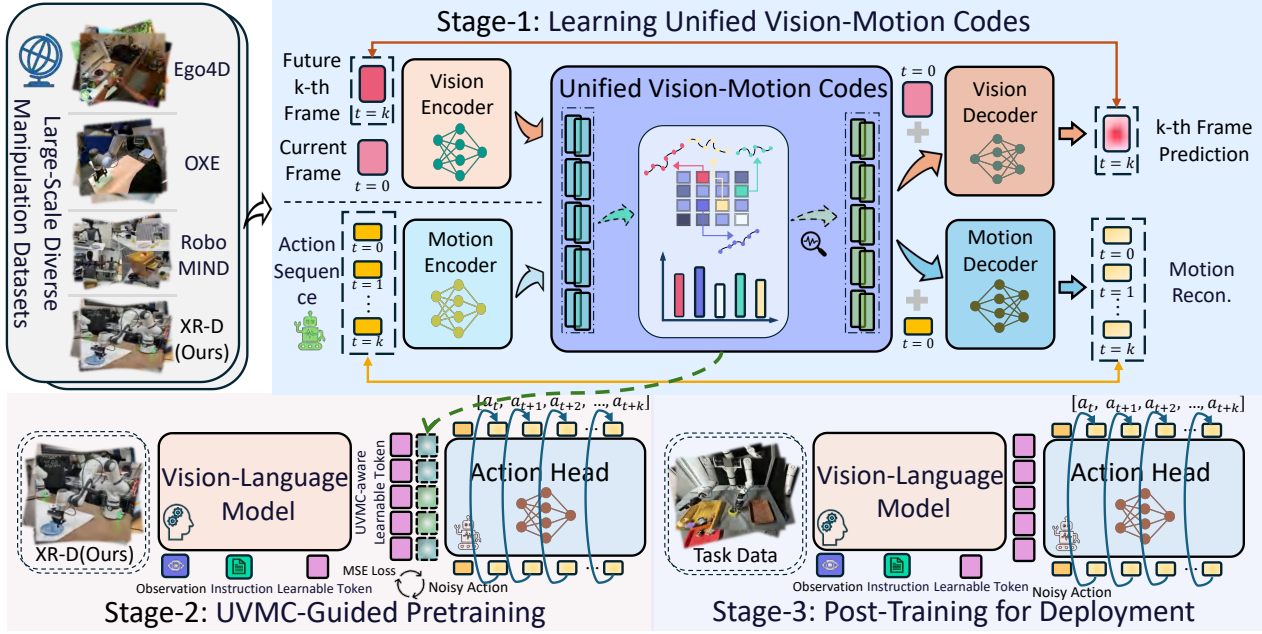


Figure 2. Overview of **X Robotic Model 1 (XR-1)**. In XR-1, we introduce the *Unified Vision-Motion Codes (UVMC)*, a discrete latent representation that jointly encodes visual dynamics and robotic motion. XR-1 adopts a three-stage training paradigm to enable precise low-level control across diverse robots and tasks.

3. Methodology

3.1. Overview

Our goal is to build a versatile and generalist Vision-Language-Action (VLA) model that controls diverse robotic embodiments across tasks. At each inference step t , the policy π receives a language instruction l and multimodal observations $o = \langle c, m \rangle$, where $c \in \mathbb{R}^{K \times 3 \times H \times W}$ denotes K RGB images from external or robot-mounted cameras, and m represents proprioceptive states. The model then predicts the next action $\hat{a} = \pi(l, o)$ in terms of joint positions and gripper commands.

We introduce **XR-1**, a scalable framework for cross-robot VLA learning (Figure 2), structured in three stages. First, we learn a dual-branch VQ-VAE to encode visual dynamics and robot motion into a unified discrete latent space, yielding *Unified Vision-Motion Codes (UVMC)*. In the second stage, these codes supervise large-scale pre-training on cross-embodiment datasets for broad generalization. Finally, the policy is fine-tuned on target embodiment data to adapt to specific dynamics and improve success rates. By progressing from the unified representation to task-specific refinement, this design ensures scalability and adaptability.

3.2. Stage-1: Learning Unified Vision-Motion Codes

We design a dual-branch VQ-VAE (Van Den Oord et al., 2017) to learn Unified Vision-Motion Codes (UVMC) via self-supervision. Unlike prior works focusing exclusively on visual dynamics (Cui et al., 2023; Hu et al., 2025; Bu

et al., 2025a; He et al., 2024; Ye et al., 2025; Cheang et al., 2025b; Du et al., 2023) or action sequences (Shafiullah et al., 2022; Wu et al., 2025b; Bauer et al., 2025; Zhang et al., 2026; Mete et al., 2024; Chen et al., 2025), our approach explicitly unifies both modalities in a discrete latent space. By employing an alignment regularization loss, our design provides complementary guidance for action prediction and enables learning from heterogeneous sources, such as actionless human demonstrations.

Visual Dynamic Code Extraction. Vision captures universal dynamics across robots and environments. To encode temporal visual variations in the vision branch, we adopt an asymmetric VQ-VAE (Zhu et al., 2023c) structure tailored for future-frame prediction. Given two frames c_t and c_{t+h} , the vision encoder $E_{\text{vis}}(\cdot)$ produces a latent code $z_{\text{vis}} = E_{\text{vis}}(c_t, c_{t+h})$, which compresses temporal changes over h steps. The decoder then predicts the future frame via $\hat{c}_{t+h} = D_{\text{vis}}(c_t, z_{\text{vis}})$. Thus, z_{vis} captures the essential visual dynamics.

Robotic Motion Extraction. The second branch encodes low-level actions and proprioceptive states. Specifically, the motion encoder $E_{\text{mo}}(\cdot)$ takes $(a_{t:t+h}, m_{t:t+h})$ as input and outputs $z_{\text{mo}} = E_{\text{mo}}(a_{t:t+h}, m_{t:t+h})$. Unlike the vision branch, no raw images or instructions are used here to ensure that the representation focuses purely on robotic dynamics. The motion decoder $D_{\text{mo}}(\cdot)$ then takes the latent motion embedding z_{mo} and optional conditions cd as input, such as the language instruction l , proprioceptive states m , and the observations o . The decoder reconstructs actions as

$\hat{\mathbf{a}}_{t:t+h} = D_{\text{mo}}(\mathbf{z}_{\text{mo}}, \mathbf{cd})$. In our implementation, we use the proprioceptive states $\mathbf{m}$ as the condition input.

Unified Vision-Motion Codes. To unify both modalities, we introduce a VQ-VAE codebook $\mathbf{e} \in \mathbb{R}^{d \times f}$ with d discrete entries of dimension f . Encoder outputs $\mathbf{z}_{\text{vis}}$ and $\mathbf{z}_{\text{mo}}$ are quantized by nearest-neighbor lookup: $\mathbf{z}_{\text{vis}}^e = S(\mathbf{z}_{\text{vis}}) = \mathbf{e}_j$, where $j = \arg \min_i \|\mathbf{z}_{\text{vis}} - \mathbf{e}_i\|_2$, and $\mathbf{z}_{\text{mo}}^e = S(\mathbf{z}_{\text{mo}}) = \mathbf{e}_j$, where $j = \arg \min_i \|\mathbf{z}_{\text{mo}} - \mathbf{e}_i\|_2$. Both decoders then condition on these quantized codes for reconstruction. Training follows standard VQ-VAE objectives (Van Den Oord et al., 2017), combining reconstruction losses with codebook and commitment regularization terms:

$$\begin{aligned} \mathcal{L}_{\text{vis}} &= \|\hat{\mathbf{c}}_{t+h} - \mathbf{c}_{t+h}\|_1 \\ &\quad + \beta \|sg(\mathbf{z}_{\text{vis}}) - \mathbf{z}_{\text{vis}}^e\|_2^2 + \beta \|\mathbf{z}_{\text{vis}} - sg(\mathbf{z}_{\text{vis}}^e)\|_2^2, \\ \mathcal{L}_{\text{mo}} &= \|\hat{\mathbf{a}}_{t:t+h} - \mathbf{a}_{t:t+h}\|_1 \\ &\quad + \beta \|sg(\mathbf{z}_{\text{mo}}) - \mathbf{z}_{\text{mo}}^e\|_2^2 + \beta \|\mathbf{z}_{\text{mo}} - sg(\mathbf{z}_{\text{mo}}^e)\|_2^2, \end{aligned} \quad (1)$$

where $sg(\cdot)$ denotes stop-gradient. We set $\beta = 0.25$ in all experiments. To capture both the vision and motion signals, we concatenate the robotic motion codes $\mathbf{z}_{\text{mo}}^e$ and visual dynamics codes $\mathbf{z}_{\text{vis}}^e$ to obtain the Unified Vision-Motion Codes $\mathbf{z}_{\text{uvmc}}^e$ for subsequent policy learning.

Cross-Modality Alignment. While motion codes provide precise control signals, visual embeddings may capture irrelevant factors (e.g., camera jitter). To mitigate this gap, we introduce an alignment loss that constrains visual codes to remain consistent with their motion counterparts:

$$\mathcal{L}_{\text{align}} = D_{\text{KL}}(q(\mathbf{z}_{\text{mo}}^e) \| q(\mathbf{z}_{\text{vis}}^e)), \quad (3)$$

where $q(\cdot)$ denotes the posterior distribution in the codebook space. This grounding of perception in motor dynamics improves robustness and allows human-only demonstrations to be effectively mapped into the robot’s action space.

Final Training Objective. The overall objective integrates reconstruction and alignment losses from different data sources. For robotic demonstrations, we jointly optimize $\mathcal{L}_{\text{total}}^{\text{robot}} = \mathcal{L}_{\text{vis}} + \mathcal{L}_{\text{mo}} + \mathcal{L}_{\text{align}}$, where $\mathcal{L}_{\text{vis}}$ and $\mathcal{L}_{\text{mo}}$ are the VQ-VAE losses for visual and motion branches, and $\mathcal{L}_{\text{align}}$ enforces cross-modal consistency. For human demonstrations, where low-level actions are unavailable, the objective naturally reduces to $\mathcal{L}_{\text{total}}^{\text{human}} = \mathcal{L}_{\text{vis}}$. This design allows training on both robot rollouts and purely visual human data. Further architectural details are provided in Appendix B.

3.3. Stage-2: UVMC-Guided Generalist Pretraining

After learning the Unified Vision-Motion Codes (UVMC) with the dual-branch VQ-VAE, we integrate it into policy learning to enhance low-level control. The policy $\pi(\cdot)$ follows a standard VLA design with a VLM $F(\cdot)$ and an action

head $H(\cdot)$. Learnable tokens $\mathbf{t}$ are introduced into the VLM input, enabling $F(\cdot)$ to predict the UVMC. The prediction loss is defined as $\mathcal{L}_{\text{uvmc}} = \|F(\mathbf{l}, \mathbf{o}, \mathbf{t}) - \mathbf{z}_{\text{uvmc}}^e\|_2^2$. In parallel, the action head is pretrained on robot datasets using an action loss $\mathcal{L}_{\text{act}}$, which may be generative or autoregressive depending on the model variant. The overall objective is $\mathcal{L} = \mathcal{L}_{\text{uvmc}} + \mathcal{L}_{\text{act}}$. This joint training encourages the backbone to internalize structured vision-motion representations while ensuring effective large-scale action pretraining.

3.4. Stage-3: Post-training for Deployment

Following large-scale UVMC-guided pre-training, the model effectively extracts unified vision-motion knowledge to produce foundation-level actions. To refine performance on downstream tasks, we introduce a post-training stage where the VLA policy is fine-tuned on task-specific datasets using an action loss $\mathcal{L}_{\text{act}}$. A primary advantage of our framework is its model-agnostic design, making it directly compatible with various VLA architectures. This flexibility allows for the integration of diverse backbones while consistently leveraging the benefits of our pre-trained representations.

3.5. Data Collection and Implementation Details

Dataset Collection. To support large-scale pretraining, we curate a comprehensive dataset by integrating four complementary sources: Open-X (O’Neill et al., 2024), RoboMIND (Wu et al., 2025a), Ego4D (first-person human activity videos) (Grauman et al., 2022), and XR-D (a subset of RoboMIND 2.0 (Hou et al., 2025) primarily comprising in-house collected data across multiple robot embodiments).

Table 1 summarizes the distribution of episodes and frames across these datasets, together with their relative proportions. Since the number of episodes and frames varies significantly among different sources, we assign dataset-specific sampling weights during training to balance contributions and prevent overfitting to dominant datasets. We provide more details of the datasets in Appendix C.

Implementation Details. The framework is model-agnostic.

Our main instantiation follows the design of π_0 (Black et al., 2025), built

on PaliGemma (Beyer et al., 2024) with a SigLIP visual encoder (Zhai et al., 2023), Gemma backbone (Team et al., 2024a), and action head. We also provide a lightweight variant, XR-1-Light, built upon SwitchVLA (Li et al., 2025a), which uses Florence-2 (Xiao et al., 2024) to reduce computational cost with minimal performance degradation. Further details on training settings and model configurations are provided in Appendix B.

Table 1. Dataset Statistics.

Dataset	Episodes	Frames	Weight
OXE	978k	59.3M	40%
RoboMIND	69k	21.4M	15%
XR-D	158k	69.1M	35%
Ego4D	59k	14.3M	10%

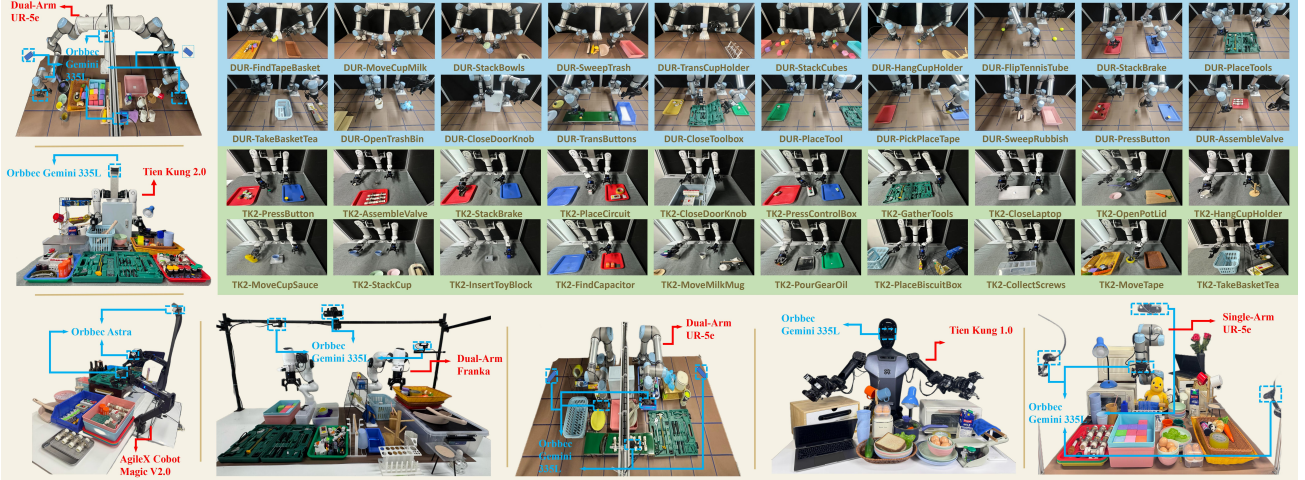


Figure 3. Experimental Setup. We evaluate XR-1 across six robot embodiments (Tien Kung 1.0/2.0, Single-/Dual-Arm UR-5e, Dual-Arm Franka, and AgileX Cobot Magic V2.0), covering more than 120 manipulation tasks with over 14k rollouts.

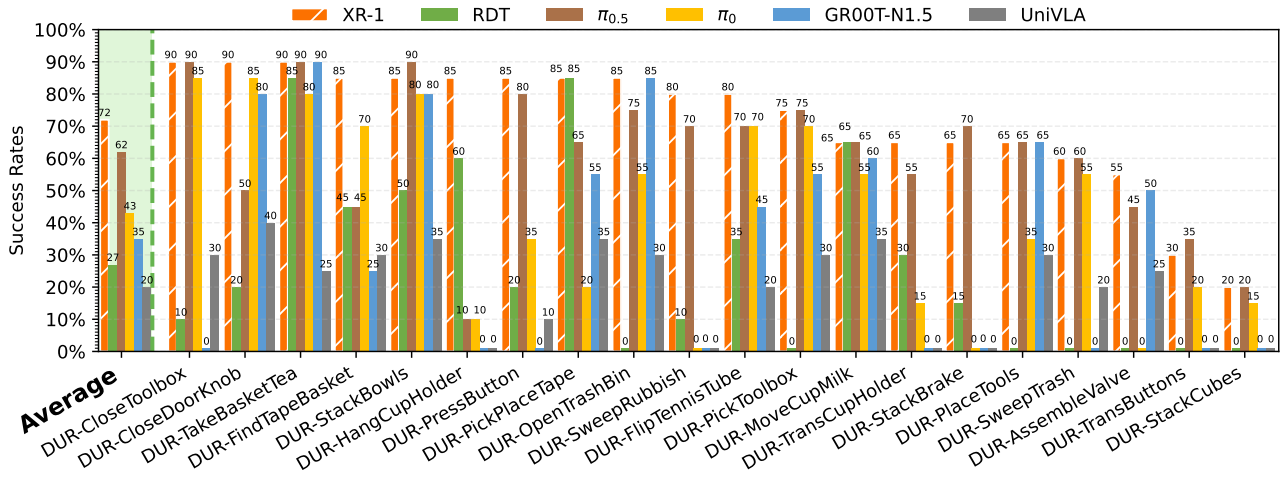


Figure 4. Success rate results across 20 tasks on Dual-Arm UR-5e.

4. Experiments

We evaluate XR-1 across six robotic embodiments and over 120 tasks, including bimanual collaboration, dexterous manipulation, and long-horizon tasks, to address four key questions: (i) performance compared to SOTA VLA models; (ii) the impact of large-scale pre-training on rapid adaptation; (iii) generalization to novel objects and environmental shifts; and (iv) the efficacy of specific components and training strategies. By benchmarking against multiple strong baselines, we demonstrate the robustness and scalability of our approach in diverse, challenging scenarios.

4.1. Experiment Setup

Real-World Robotic Setup. We evaluate XR-1 on six heterogeneous embodiments (Figure 3): Tien Kung 1.0/2.0, single/dual-arm UR-5e, dual-arm Franka, and AgileX Cobot Magic V2.0. Each platform is equipped with parallel grippers

and calibrated multi-view cameras. For each embodiment, 20 tasks were designed and expert demonstrations were collected via teleoperation, recording synchronized RGB and proprioceptive streams (e.g., joint positions and gripper commands). Representative tasks for the Dual-Arm UR-5e and Tien Kung 2.0 are shown in Figure 3, with full task details provided in Appendix K.

Training and Evaluation Protocol. We employ a three-stage pipeline: (i) UVMC Pre-training: learning representations from large-scale heterogeneous datasets, including RoboMIND (Wu et al., 2025a), Open-X (O’Neill et al., 2024), XR-D, and Ego4D (Grauman et al., 2022); (ii) Policy Pre-training: integrating cross-embodiment knowledge on XR-D; and (iii) Task Fine-tuning: refining performance on target tasks. This staged training progressively transfers vision-motion abstractions to downstream robotic policies. For evaluation, we conduct 20 rollouts per task and report success rates based on human evaluation.

Table 2. Success rate results across 20 tasks on Tien Kung 2.0.

Method	TK2-Press Button	TK2-Assemble Valve	TK2-Stack Brake	TK2-Place Circuit	TK2-Press ControlBox	TK2-Close DoorKnob	TK2-Gather Tools	TK2-Close Laptop	TK2-Open PotLid	TK2-Hang CupHolder	-
UniVLA	25	0	25	50	10	0	0	20	35	0	-
RDT	15	0	0	65	20	0	0	0	90	0	-
GR00T-N1.5	85	20	85	90	0	0	20	0	75	0	-
π_0	85	10	55	70	85	0	20	85	85	20	-
$\pi_{0.5}$	80	0	65	60	40	25	20	90	80	35	-
XR-1 (ours)	90	15	90	90	90	85	25	90	85	75	-
	TK2-Move CupSauce	TK2-Stack Cup	TK2-Insert ToyBlock	TK2-Find Capacitor	TK2-Move MilkMug	TK2-Pour GearOil	TK2-Place BiscuitBox	TK2-Collect Screws	TK2-Move Tape	TK2-Take BasketTea	Avg.
UniVLA	45	30	0	25	35	35	0	0	20	0	17.8
RDT	50	25	0	0	0	75	0	0	0	0	17.0
GR00T-N1.5	50	55	0	60	55	70	25	0	70	0	38.0
π_0	60	20	10	0	80	55	0	0	0	75	40.8
$\pi_{0.5}$	70	25	20	0	75	75	0	0	0	60	41.0
XR-1 (ours)	70	85	55	75	90	85	75	15	70	85	72.0

Table 3. Ablation study of XR-1. In Stage-1 and Stage-2, “DT” indicates training directly on the downstream task data.

Exp.	Instantiation	Stage-1	Stage-2	Stage-3	DUR-Clean Table	DUR-Find TapeBasket	DUR-Move CupMilk	DUR-Stack Bowls	DUR-Sweep Trash	DUR-Trans CupHolder	Avg
1	XR-1-Light	×	×	✓	0	70	0	75	60	50	42.5
2	XR-1-Light	DT	DT	✓	40	90	10	90	60	55	57.5
3	XR-1	×	×	✓	0	50	20	55	0	45	28.3
4	XR-1 w/o KL	DT	DT	✓	45	55	35	60	30	65	48.3
5	XR-1	DT	DT	✓	50	75	65	80	60	70	66.7
6	XR-1	1%	DT	✓	15	60	10	55	15	20	29.2
7	XR-1	10%	DT	✓	25	60	25	60	20	40	38.3
8	XR-1	50%	DT	✓	25	80	65	80	20	50	53.3
9	XR-1	100%	DT	✓	60	80	70	85	40	55	65.0
10	XR-1	100%	XR-D	✓	70	85	80	90	85	80	81.6

4.2. Results on Real-World Robotic Tasks

Baseline Methods. We compare XR-1 with strong VLA models, including $\pi_{0.5}$ (Intelligence et al., 2025), π_0 (Black et al., 2025), RDT (Liu et al., 2025), UniVLA (Bu et al., 2025b), and GR00T-N1.5 (Bjorck et al., 2025). We note a performance degradation with the Lerobot implementation of π_0 . The results of π_0 reported in this paper are based on the original JAX implementation.

Results on Dual-Arm UR-5e. Figure 4 shows success rates on 20 Dual-Arm UR-5e tasks. XR-1 outperforms all baselines by a wide margin (e.g., *DUR-FindTapeBasket*: 85% vs. 50% for π_0), while several baselines drop to 0% on harder tasks. We attribute this to limited auxiliary supervision and optimization conflicts in multi-task training. XR-1 mitigates these issues with UVMC, producing stronger representations and more stable optimization. Tabular results are in Appendix Table 13.

Results on Tien Kung 2.0 (Unseen during Pretraining). We further evaluate transferability on Tien Kung 2.0 over another 20 tasks in Table 2. Unlike the UR-5e, this robot is *unseen during pretraining* (e.g., Stages 1 and 2 for XR-1), making the evaluation a stringent embodiment-transfer benchmark. Despite this challenge, XR-1 again outperforms all baselines; e.g., in *TK2-MoveCupSauce*, it reaches 70% versus 60% for π_0 . These results indicate that UVMC effec-

tively encodes embodiment-agnostic dynamics into a shared latent space, enabling efficient transfer of prior knowledge to novel robotic platforms.

Results on Other Robots. XR-1 consistently outperforms all other methods across four diverse robotic arm configurations, achieving a significant relative gain over the strongest baseline. Additional experimental results for Tien Kung 1.0 in Table 9, Dual-Arm Franka in Table 10, AgileX Cobot Magic V2.0 in Table 11, and Single-Arm UR-5e in Table 12 are provided in Appendix D.

4.3. Ablation Study

To disentangle the contribution of each component in XR-1, we conduct ablations on six manipulation tasks using the Dual-Arm UR-5e. Table 3 summarizes success rates under different configurations, covering model capacity, UVMC learning, cross-modal alignment, and dataset scaling. Additional experimental results and analysis for the UVMC ablation study, as well as for Ego4D and cross-embodied knowledge transfer ablations on enhanced single-embodiment performance, are provided in Appendix E.

Lightweight Models. To validate the applicability of our methods in resource-constrained environments, we extend our evaluation to a compact variant, XR-1-Light, which comprises only 230M trainable parameters. When this model

Table 4. Generalization results of XR-1 on unseen scenarios.

Method	DFR-SweepTrash			DFR-HangCup		
	Novel Objects (rubbish)	Novel Objects (dustpan)	Dynamic Distractors	Background Variations	Illumination Changes	Static Distractors
π_0	15	50	5	30	15	10
XR-1 (ours)	65	60	55	55	30	30

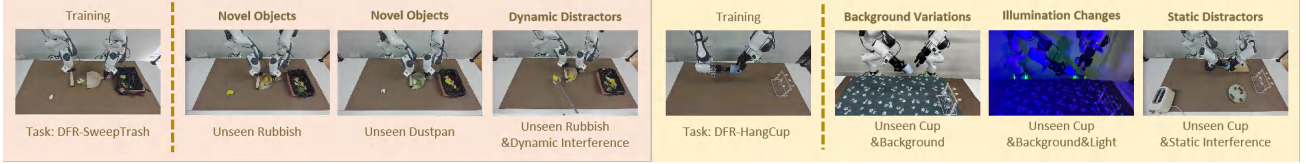


Figure 5. Unseen scenario task setup on Dual-Arm Franka.

is trained without our proposed UVMC-based fine-tuning stage (Exp. 1), it achieves a respectable baseline performance of 42.5%. However, the integration of the UVMC module during its training phase (Exp. 2) elevates the average success rate to 57.5%. *This significant +15% absolute improvement confirms that our training methodology is not merely an artifact of large model capacity but provides substantial and consistent benefits even for highly efficient, lightweight architectures, thereby highlighting its broad applicability and practical value.*

UVMC and Cross-Modal Alignment. Exps. 3–5 examine the role of UVMC together with a cross-modal alignment loss between vision and motion. Performance consistently improves as these components are added, confirming their complementary importance for feature learning across tasks.

Scaling with Pretraining Data. Having established the profound impact of our core training objectives, we now turn our attention to the role of large-scale pretraining. We first analyze the scaling behavior with respect to the volume of Stage-1 pretraining data, using the full XR-1 model without any subsequent fine-tuning. The results from Exps. 6–9 reveal a clear and compelling monotonic scaling law: as the pretraining dataset size systematically increases from 1% to 10%, 50%, and finally 100%, the average success rate consistently climbs from 29.2% to 38.3%, 53.3%, and ultimately 65.0%. *This trend unequivocally demonstrates the foundational importance of leveraging large and diverse datasets to learn generalizable robotic policies.* Intriguingly, the performance achieved with 100% of the Stage-1 pretraining data (65.0%) is already on par with the model that was exclusively fine-tuned on downstream tasks (Exp. 5, 66.7%), indicating that task-agnostic pretraining can effectively instill a powerful inductive bias.

4.4. Generalization Analysis

Generalization to Unseen Scenarios. We further evaluate XR-1 on unseen conditions to assess its out-of-distribution generalization. As shown in Figure 5, we test on (i) novel

objects (e.g., unseen rubbish or dustpans), (ii) dynamic and static distractors, (iii) illumination changes, and (iv) background variations. As shown in Table 4, XR-1 consistently outperforms the strong VLA method π_0 across all settings. It demonstrates clear gains on novel objects, improved robustness under distractor interference, and stable performance when background and lighting variations are introduced. *These results highlight XR-1’s strong generalization not only across embodiments and tasks but also under diverse environmental shifts never encountered during pretraining or fine-tuning, underscoring its potential for real-world deployment.*

Out-of-Box Evaluation. We assess the foundational capabilities of XR-1 after Stage-2, bypassing Stage-3 post-training. We evaluate on 7 tasks from the Dual-Arm Franka setup, which comprises only 0.45% of the XR-D dataset. For a fair comparison, baselines without XR-D pre-training are fine-tuned on these specific tasks. As shown in Figure 6, the pre-trained XR-1-oob model, despite no adaptation, achieves performance comparable to GR00T-N1.5 and π_0 , while outperforming RDT and UniVLA. *This robustness highlights the efficacy of UVMC in aligning multimodal dynamics across embodiments, enabling strong generalization with minimal task-specific supervision.* Additional results for the Dual-Arm UR-5e are in Appendix F.

Fast Adaptation to New Tasks. We further evaluate whether XR-1 can rapidly adapt to unseen tasks with limited demonstrations. Specifically, we collect 15 new tasks on Tien Kung 2.0 (unseen in XR-D), each with 20 trajectories. XR-1 is trained jointly across these tasks, while single-task baselines, ACT (Zhao et al., 2023) and Diffusion Policy (DP) (Chi et al., 2023), are trained independently per task. As shown in Figure 7, XR-1 achieves significantly higher success rates than ACT and DP, despite the setting favoring the baselines. *This advantage stems from large-scale pre-training and UVMC supervision, enabling XR-1 to extract transferable features from few-shot data and adapt across diverse embodiments.* Additional results and analysis on the Dual-Arm UR-5e are provided in Appendix F.

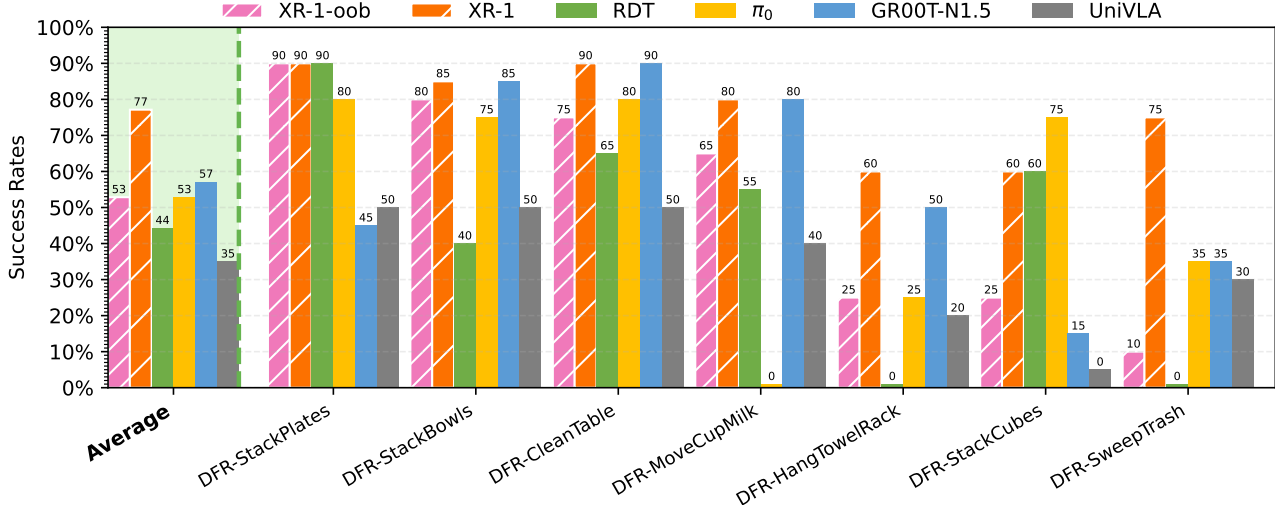


Figure 6. Out-of-box evaluation results of 7 tasks on Dual-Arm Franka.

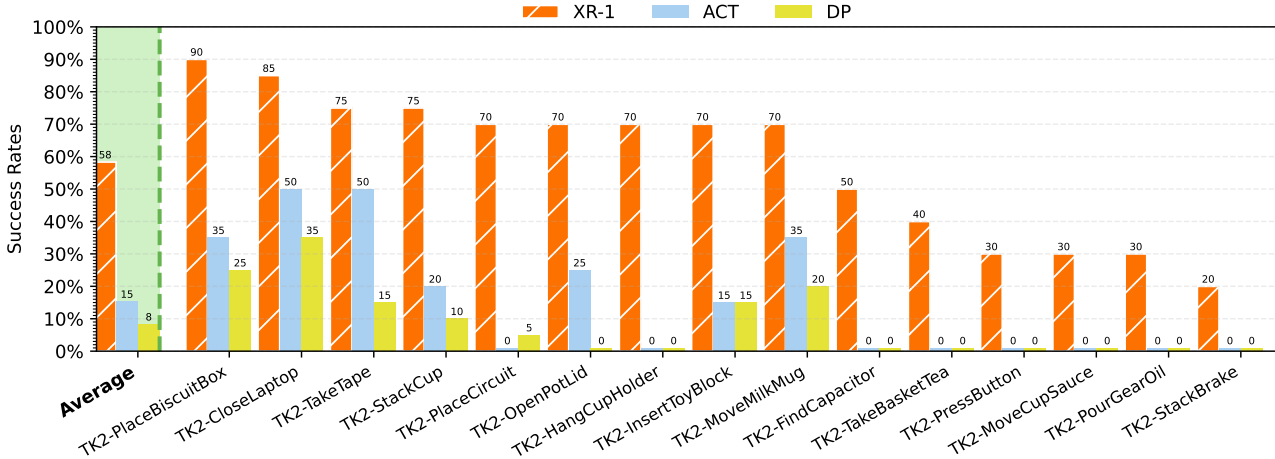


Figure 7. Fast adaptation on Tien Kung 2.0. Tien Kung 2.0 is an unseen embodiment in XR-D. In this setup, XR-1 adapts to 15 novel tasks with one model using only 20-shot demonstrations per task, while baselines (ACT and DP) are trained per task.

4.5. Additional Analyses

Additional simulation results are provided in Appendix G. To assess semantic alignment between visual codes (VC) and motion codes (MC), we perform cross-modal nearest-neighbor retrieval and t-SNE visualization. These analyses validate UVMC’s ability to capture embodiment-agnostic action semantics: nearest-neighbor retrieval shows that MCs retrieve visually consistent action phases across tasks and embodiments, while t-SNE reveals coherent clustering by task dynamics rather than robot morphology. Failure analyses for baselines and XR-1 are provided in Appendix I and Appendix J, respectively, showing that XR-1 reduces baseline failures such as optimization collapse, conflicting gradients, and coordination errors, though high-precision manipulation remains challenging. We further categorize the 120 real-world tasks into seven types in Appendix K. The full benchmark specification, including task images, language instructions, and collected demonstration counts for all 120 tasks, is provided in Appendix K.1.

5. Conclusion

We presented **X Robotic Model 1 (XR-1)**, a unified framework for versatile and scalable vision-language-action learning that addresses the key limitations of existing approaches: precise low-level action generation and cross-domain multimodal knowledge exploitation across heterogeneous data sources. Central to our approach is the *Unified Vision-Motion Codes (UVMC)*, which serve as embodiment-agnostic abstractions aligning visual dynamics with motor control through a shared discrete latent space. By utilizing a three-stage training paradigm, XR-1 achieves robust performance across diverse robots, environments, and tasks while significantly outperforming state-of-the-art baselines such as $\pi_{0.5}$, π_0 , RDT, UniVLA, and GR00T-N1.5. Our results highlight the importance of multimodal alignment for embodied AI and suggest promising directions toward general-purpose robotic agents capable of interacting with the physical world and adapting seamlessly to new environments and task demands.

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Developing large-scale Vision-Language-Action (VLA) models is a monumental task that required extensive collaboration among numerous researchers across multiple domains. The development of this work would not have been possible without the dedication and expertise of many individuals who contributed their time and knowledge throughout various stages of the project.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Open-Source Resources

- Code Repository: <https://github.com/Open-X-Humanoid/XR-1>.
- Model Weights (HuggingFace): <https://huggingface.co/collections/X-Humanoid/xr-1>.
- Model Weights (ModelScope): <https://modelscope.cn/collections/X-Humanoid/XR-1>.

B. Implementation Details

In this section, we provide a detailed description of the XR-1 framework, focusing on the architecture and training of the dual-branch VQ-VAE. The model is designed to encode both vision dynamics and robotic motion into a shared discrete latent space, thereby enabling seamless integration of perception and control.

B.1. Dual-Branch VQ-VAE

To achieve a unified latent representation, we introduce a dual-branch VQ-VAE consisting of two complementary encoders, a vision encoder and a motion encoder, that map their respective modalities into a common discrete codebook. Each branch is paired with a decoder to facilitate reconstruction during pretraining. The overall design ensures that the majority of representational capacity resides in the encoders, while the decoders primarily serve as auxiliary components for reconstruction.

Vision Branch. The vision branch processes raw image observations $\{o_t, o_{t+h}\}$ and encodes them into compact latent tokens.

Vision Branch Encoder. We adopt SigLIP (Zhai et al., 2023) as the backbone vision encoder, comprising approximately 400M parameters. This encoder extracts high-level features from visual inputs. To capture temporal dynamics beyond static representations, we incorporate a visual dynamic module inspired by (Chen et al., 2025). This module is implemented as a four-layer transformer (ViT (He et al., 2022)) with 32M parameters, which compresses vision dynamic information into a fixed number of latent tokens by querying dynamic features.

Vision Branch Decoder. For reconstruction, we employ a ViT-based decoder with 12 transformer layers (94M parameters). Importantly, the encoder contains roughly five times more parameters than the decoder. This asymmetry is intentional: by allocating more capacity to encoding, we encourage the model to produce informative latent tokens that simplify downstream decoding. Consequently, the decoder remains lightweight since its role is auxiliary rather than representationally dominant.

During training, all parameters in both the SigLIP backbone and the dynamic module remain fully trainable. Additional details regarding training hyperparameters are provided in Table 5.

Motion Branch. The motion branch encodes action sequences $\{a_{t:t+h}\}$ into discrete motion codes.

Motion Branch Encoder. To capture temporal dependencies across actions, we employ 1D causal strided convolutions (Van Den Oord et al., 2016), which progressively reduce sequence length h while preserving causality. The stride configuration determines the degree of temporal abstraction achieved at each stage. Following this convolutional compression, an 8-layer transformer encoder (34M parameters) further contextualizes action embeddings before quantization into discrete tokens.

Motion Branch Decoder. For action reconstruction, we leverage Gemma (Beyer et al., 2024), an autoregressive language model with approximately 300M parameters. The design closely follows the action expert structure in π_0 (Black et al., 2025), integrating diffusion-based supervision for reconstructing low-level actions from motion codes. Pretraining this decoder equips it with strong generative priors over action sequences, thereby providing an effective initialization for downstream policy learning. Additional details regarding training hyperparameters are provided in Table 5.

Overall, this dual-branch architecture ensures that both perception and motion are represented in a unified tokenized space via vector quantization (VQ), enabling scalable pretraining across multimodal data sources.

B.2. XR-1 Models

XR-1. The proposed framework is designed to be model-agnostic, making it compatible with a wide range of vision-language-action (VLA) architectures. In this work, we instantiate XR-1 using a configuration inspired by the baseline policy

Table 5. Implementation Details of Dual-Branch VQ-VAE.

	Hyperparameter	Value		Hyperparameter	Type	Params.
Hyper-parameter	Batch Size	960	Network Architectures	Vision Encoder	SigLIP	400M
	Learning Rate	1e-4		Vision Dynamic Encoder	ViT	32M
	Optimizer	AdamW		Vision Decoder	ViT	94M
	Trainable Parameters	0.9B		Vision Recons. Loss	MSE	-
	Motion/Vision Codebook Category	256		Action Encoder	Convolution and Transformer	33M
	Motion/Vision Codebook Embed. Dim	256		Action Decoder	Transformer Decoder	300M
	Motion/Vision Code Num.	13		Action Recons. Loss	Flow Matching	-
	Action Sequence	50		-	-	-
	Vision Interval	50		-	-	-
	Training Step	275K		-	-	-

π_0 (Black et al., 2025) while introducing several key modifications that enable more structured representation learning. Specifically, XR-1 builds upon the PaliGemma architecture (Beyer et al., 2024), which integrates a SigLIP-based visual encoder (Zhai et al., 2023) with approximately 400 million parameters and a Gemma transformer backbone (Team et al., 2024a) with an action prediction head containing around 2.6 billion parameters. This design largely mirrors π_0 in terms of scale and backbone selection, but diverges in how supervision is introduced.

Instead of directly optimizing for action prediction as in π_0 , XR-1 leverages the UVMC produced by a Dual-Branch VQ-VAE as intermediate supervisory signals. The joint representation z_{umvc} encodes both motion and visual dynamics information, which serves as guidance for training. To incorporate this signal effectively, we introduce two learnable tokens, $[ZMO]$ and $[ZVIS]$, that are responsible for predicting the robotic motion codes and the visual dynamics codes. These predictions are optimized using mean squared error loss against their respective targets. By enforcing this disentangled supervision on both motor control and perceptual dynamics, XR-1 encourages stronger alignment between perception and action.

To ensure fairness in evaluation, XR-1 is initialized from PaliGemma’s publicly available pretrained checkpoint rather than directly adopting the released weights of π_0 . This avoids potential confounding effects due to differences in pretraining objectives or data exposure. Overall, XR-1 extends beyond π_0 by introducing structured supervision through VQ-VAE latent codes and dedicated learnable tokens for motion and visual prediction, while maintaining compatibility with large-scale pretrained models such as PaliGemma. Additional details regarding training hyperparameters are provided in Table 6.

Table 6. Implementation Details of XR-1.

	Hyperparameter	Value		Hyperparameter	Value
Hyper-parameter	Batch size	640	Network Architectures	Decoder layer	18
	Learning rate	1e-4		Transformer hidden dim	2048
	Optimizer	AdamW		Heads num	8
	$[ZMO]$ Number	13		Action Decoder layer	18
	$[ZVIS]$ Number	$13 * view_{num}$		Action Transformer hidden dim	1024
	Action sequence	50		Action Heads num	8
	Training step	300k		Action loss	flow matching

XR-1-Light. To further highlight the flexibility of our approach, we introduce **XR-1-Light**, a lightweight variant of XR-1 that significantly reduces computational cost while maintaining competitive performance. The motivation behind XR-1-Light is to replace the large-scale PaliGemma backbone, which contains nearly 3 billion parameters, with a more efficient vision-language model (VLM) without sacrificing the ability to capture rich multimodal representations. For this purpose, we adopt Florence-2 (Xiao et al., 2024), a transformer-based model with approximately 230 million parameters, as the backbone within the SwitchVLA framework (Li et al., 2025a). This substitution enables faster training and inference while lowering memory requirements, making XR-1-Light more suitable for resource-constrained scenarios.

Despite its reduced scale, XR-1-Light preserves the core design principles of XR-1. In particular, it continues to leverage the supervisory signal UVMC from the Dual-Branch VQ-VAE, which encodes both robotic motion and visual dynamics. To integrate this supervision effectively, we employ two learnable tokens, $[ZMO]$ and $[ZVIS]$, that are responsible for predicting the motion codes and the visual dynamics codes. Unlike in XR-1 where these tokens are attached to a decoder-only

transformer backbone, in Florence-2 they are inserted between the encoder and decoder layers. This design allows the encoder to specialize in extracting structured latent representations aligned with UVMC, while enabling the decoder to function as an action expert that generates task-specific predictions conditioned on these learned codes.

A notable difference between XR-1 and XR-1-Light lies in their training strategies. While XR-1 benefits from pretraining on XR-D before fine-tuning on downstream tasks, XR-1-Light omits this stage due to its lightweight architecture. Instead, it is directly fine-tuned on task-specific datasets. This choice reflects a trade-off: although pretraining could potentially enhance generalization, direct fine-tuning allows us to fully exploit Florence-2’s efficiency without incurring additional computational overhead.

In summary, XR-1-Light demonstrates that our framework can be instantiated not only with large-scale backbones such as PaliGemma but also with compact VLMs like Florence-2. By maintaining structured supervision through z_{uvmc} while reducing parameter count by more than an order of magnitude, XR-1-Light provides a practical alternative that balances performance with efficiency. Additional details regarding training hyperparameters are provided in Table 7.

Table 7. Implementation Details of XR-1-Light

	Hyperparameter	Value		Hyperparameter	Value
Hyper-parameter	Batch Size	160	Network Architectures	Encoder Layer	6
	Learning Rate	5e-5		Transformer Hidden Dim.	768
	Optimizer	AdamW		Heads Num.	12
	[ZMO] Number	13		Action Decoder Layer	6
	[ZVIS] Number	13* $view_{num}$		Action Transformer Hidden Dim.	768
	Action Sequence	50		Action Heads Num.	12
	Training Step	50K		Action Loss	Flow Matching

B.3. Training and Inference

The training of our framework is organized into three stages: UVMC learning, UVMC-guided pretraining, and policy fine-tuning. Each stage progressively aligns perception, representation, and control while balancing computational efficiency.

In the first stage, the UVMC module, containing approximately 0.9B parameters, is pretrained on large-scale multimodal data. This process consumed roughly 38,400 GPU hours on a cluster of 80 NVIDIA A100 GPUs (80GB each), enabling the model to capture both motion and visual dynamics representations.

The second stage involves policy pretraining, where the complete model scales up to about 4B parameters. This step also required around 38,400 GPU hours on the same hardware configuration. The objective here is to integrate the pretrained UVMC representations into a unified vision-language-action policy.

In the final stage, policy fine-tuning is performed for embodiment-specific adaptation. Each embodiment configuration is fine-tuned across 20 downstream tasks using 8 A100 GPUs (80GB), requiring approximately 576 GPU hours per embodiment. This ensures that XR-1 and its variants generalize effectively to diverse robotic environments while remaining computationally practical.

For inference, we emphasize both responsiveness and throughput. The system operates with an action chunk inference frequency of about 5 Hz while maintaining an average action-level inference rate close to 200 Hz (actions per second). These frequencies are achieved on a single commercially available RTX 4090 GPU (24GB), demonstrating that despite large-scale pretraining costs, deployment remains efficient without reliance on massive compute resources.

C. Dataset Curation

Large-scale pretraining has consistently been shown to enhance both generalization and rapid adaptation in multimodal learning systems. Motivated by these findings, we curate a comprehensive dataset tailored for robotic manipulation, integrating diverse sources of visual, linguistic, and action-centric data. Our dataset construction draws from four complementary resources: *Open-X* (O’Neill et al., 2024), which provides large-scale open-world manipulation trajectories; *RoboMIND* (Wu et al., 2025a), a benchmark emphasizing reasoning-driven robotic tasks; *Ego4D* (Grauman et al., 2022), a first-person human activity dataset offering rich egocentric perspectives; and *XR-D*, our in-house collection spanning multiple robotic

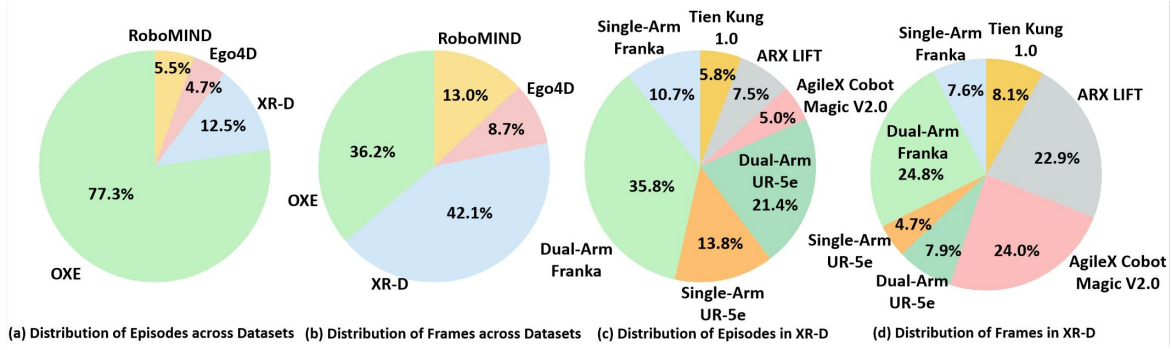


Figure 8. Overview of the pretraining datasets used for XR-1. We combine Open-X, RoboMIND, Ego4D, and our dataset XR-D, with a total of $\sim 1,264$ k episodes and 110M frames.

embodiments and task domains. Together, these sources cover a wide spectrum of sensory modalities, embodiment variations, and task complexities, forming a foundation for scalable pretraining.

The training procedure is organized into three progressive stages. In **Stage-1**, we pretrain a dual-branch VQ-VAE on the combined datasets to learn disentangled latent representations of motion and visual dynamics. In **Stage-2**, we leverage XR-D to pretrain the vision-language-action (VLA) backbone, aligning multimodal perception with action generation across diverse embodiments. Finally, in **Stage-3**, we fine-tune on novel scenes and previously unseen tasks outside XR-D in order to rigorously assess transferability and generalization beyond the pretraining distribution.

A detailed breakdown of dataset statistics, including scale, modality coverage, and embodiment diversity across all sources used for UVMC pretraining, is provided in Figure 8 and Table 8.

Table 8. Pretraining Dataset Details.

Dataset	Episode	Frames	Weight
OXE (O’Neill et al., 2024)	978,582	59.3M	40%
FMB Dataset (Luo et al., 2025)	8611	1137340	0.88%
DROID (Khazatsky et al., 2024)	92233	27044326	9.43%
Language Table (Lynch et al., 2023)	442226	7045476	45.19%
Berkeley Autolab UR5 (Chen et al.)	896	87783	0.09%
Berkeley Fanuc Manipulation (Zhu et al., 2023a)	415	62613	0.04%
Berkeley Cable Routing (Luo et al., 2024)	1482	38240	0.15%
Berkeley Gnm Cory Hall (Kahn et al., 2018)	7331	156012	0.75%
Berkeley Gnm Recon (Shah et al., 2021)	11834	610907	1.21%
Berkeley Gnm Sac Son (Hirose et al., 2023)	2955	241059	0.30%
Berkeley MVP (Radosavovic et al., 2023b)	480	45308	0.05%
Berkeley RPT (Radosavovic et al., 2023a)	908	392578	0.10%
Bridge (Ebert et al., 2022; Walke et al., 2023)	25460	813372	2.60%
BC-Z (Jang et al., 2022)	43264	6015535	4.42%
Taco Play (Rosete-Beas et al., 2023; Mees et al., 2023)	3242	213972	0.33%
NYU Franka Play Dataset (Cui et al., 2023)	365	34448	0.04%
Asu Table Top (Zhou et al., 2022b; 2023)	110	26113	0.01%
Austin Buds Dataset (Zhu et al., 2022)	50	34112	0.01%
Austin Sailor Dataset (Nasiriany et al., 2022)	240	353094	0.02%
Austin Sirius Dataset (Liu et al., 2022)	559	279939	0.05%
CMU Play Fusion (Chen et al., 2023)	576	235922	0.05%
CMU Stretch (Bahl et al., 2023; Mendonca et al., 2023)	135	25016	0.01%
Columbia Cairlab Pusht Real (Chi et al., 2023)	122	24924	0.01%

Continued on next page

Dataset	Episode	Frames	Weight
DLR EDAN Shared Control (Vogel et al., 2020; Quere et al., 2020)	104	8928	0.01%
DLR Sara Grid Clamp (Padalkar et al., 2023a)	107	7622	0.01%
DLR Sara Pour (Padalkar et al., 2023b)	100	12971	0.01%
DobbE (Shafiullah et al., 2023)	5208	1139911	0.52%
Stanford Hydra Dataset (Belkhale et al., 2023)	570	358234	0.06%
Tokyo U Lsmo (Osa, 2022)	50	11925	0.01%
Toto (Zhou et al., 2022a)	902	294139	0.10%
UCSD Kitchen Dataset (Yan & Wang, 2023)	150	3970	0.02%
UCSD Pick and Place Dataset (Feng et al., 2023)	1355	67750	0.14%
UTAustin Mutex (Shah et al., 2023)	1500	361883	0.15%
U-Tokyo PR2 Opening Fridge (Oh et al., 2023)	64	9140	0.01%
U-Tokyo PR2 Tabletop Manipulation (Oh et al., 2023)	192	26346	0.02%
U-Tokyo xArm Bimanual (Oh et al., 2023)	64	1388	0.01%
U-Tokyo xArm Pick and Place (Oh et al., 2023)	92	6789	0.01%
Viola (Zhu et al., 2023b)	135	68913	0.01%
Fractal (Brohan et al., 2023)	87212	3786400	8.91 %
Furniture Bench Dataset (Heo et al., 2023)	5100	3948057	0.51%
IAMLab CMU Pickup Insert (Saxena et al., 2023)	631	146241	0.06%
Jaco Play (Dass et al., 2023)	976	70127	0.10%
Kaist Non-prehensile (Kim et al., 2023)	201	32429	0.02%
Kuka (Kalashnikov et al., 2018)	209880	2455879	21.45%
NYU Door Opening Surprising Effectiveness (Pari et al., 2022)	435	18196	0.04%
NYU ROT Dataset (Haldar et al., 2023)	14	440	0.01%
RoboSet (Bharadhwaj et al., 2024)	18250	1419999	1.86%
Roboturk (Mandlekar et al., 2019)	1796	168423	0.18%
RoboMIND (Wu et al., 2025a)	69274	21.4M	15 %
Single-Arm Franka	16018	2268033	23.12%
Dual-Arm Franka	1774	375807	2.56%
Single-Arm UR-5e	25721	2643322	37.13%
AgileX Cobot Magic V2.0	10059	6477564	14.52%
Tien Kung 1.0	15702	9683213	22.67%
XR-D	158639	69.1M	35 %
Single-Arm Franka	16933	5240845	10.67%
Dual-Arm Franka	56800	17140497	35.80%
Single-Arm UR-5e	21954	3218116	13.84%
Dual-Arm UR-5e	33916	5463729	21.38%
AgileX Cobot Magic V2.0	8004	16576019	5.05%
ARX LIFT	11866	15845836	7.48%
Tien Kung 1.0	9166	5605573	5.78%
Ego4D (Grauman et al., 2022)	59427	14.3M	10 %

D. Additional Real-World Experiments

Table 9. Success rate results across 20 tasks on Tien Kung 1.0.

Method	TK1-Close Drawer	TK1-Flip TennisTube	TK1-Press CookerButton	TK1-Move ChopstickCup	TK1-Stack Cubes	TK1-Stack Cups	TK1-Stack Plates	TK1-Pick WipeTowel	TK1-Hang Towel	TK1-Open PotLid	-
UniVLA	25	0	25	10	0	0	35	0	0	0	-
RDT	45	0	65	0	0	0	70	0	0	0	-
GR00T-N1.5	75	20	85	20	0	0	70	0	0	0	-
π_0	75	40	45	25	0	0	80	0	0	0	-
$\pi_{0.5}$	75	35	55	60	0	0	75	0	0	0	-
XR-1 (ours)	80	50	90	65	65	20	85	55	65	20	-
	TK1-Open Oven	TK1-Pack EggBox	TK1-Close Laptop	TK1-Insert Toaster	TK1-Flip Cup	TK1-Place FlipButton	TK1-Open TrashBin	TK1-Press Machine	TK1-Find Tape	TK1-Stack Bowls	Avg.
UniVLA	30	0	0	30	0	0	25	20	20	30	12.5
RDT	0	0	90	0	0	0	85	55	0	0	20.5
GR00T-N1.5	55	0	15	40	0	0	70	45	45	45	29.3
π_0	75	10	90	65	10	15	75	70	70	80	41.3
$\pi_{0.5}$	80	20	95	65	30	25	65	75	75	80	45.5
XR-1 (ours)	80	65	95	70	65	65	85	75	75	90	68.0

Results on Tien Kung 1.0. Table 9 reports success rates across 20 tasks on Tien Kung 1.0. XR-1 again outperforms all baselines by a clear margin. For example, in *TK1-HangTowel*, it achieves 65% success while all baselines fail (0%). Overall, XR-1 attains an average success rate of 68.0%, substantially higher than π_0 (41.3%) and more than double RDT (20.5%) and UniVLA (12.5%). These results highlight the effectiveness of UVMC supervision in providing robust representations and stable optimization across diverse manipulation skills.

Table 10. Success rate results across 20 tasks on Dual-Arm Franka.

Method	DFR-Move CupMilk	DFR-Stack Bowls	DFR-Sweep Trash	DFR-Transfer Cup	DFR-Move Chopstick	DFR-Stack Cubes	DFR-Stack Plates	DFR-Clean Table	DFR-Hang CupHolder	DFR-Hang TowelRack	-
UniVLA	15	20	0	30	0	5	25	35	0	20	-
RDT	55	40	0	0	15	60	90	65	0	0	-
GR00T-N1.5	80	85	35	55	0	15	45	90	25	50	-
π_0	0	75	35	60	0	75	80	80	0	25	-
$\pi_{0.5}$	15	85	60	40	0	55	90	85	45	20	-
XR-1 (ours)	80	85	75	90	55	60	90	90	65	60	-
	DFR-Find TapeBox	DFR-Pick ButtonPress	DFR-Sweep Rubbish	DFR-Close Toolbox	DFR-Collect BasketTea	DFR-Place Tools	DFR-Get Blocks	DFR-Place RagWipe	DFR-Open Toolbox	DFR-Place Screws	Avg.
UniVLA	25	0	0	15	10	0	25	20	0	0	12.3
RDT	0	20	0	25	0	0	5	55	0	0	21.5
GR00T-N1.5	80	0	0	85	35	0	70	0	0	0	37.5
π_0	90	30	0	0	0	0	75	70	50	0	37.3
$\pi_{0.5}$	65	25	30	0	0	60	75	70	0	0	41.0
XR-1 (ours)	90	75	60	90	75	60	85	70	60	55	73.5

Results on Dual-Arm Franka. Table 10 reports success rates across 20 tasks on the Dual-Arm Franka. XR-1 achieves the highest average performance (73.5%), substantially outperforming π_0 (37.3%) and other baselines. For example, in *DFR-TransferCup* it reaches 90% success, while all alternatives fall below 60%. It is because XR-1 leverages UVMC for richer supervision, yielding robust representations and stable learning across diverse objectives.

Table 11. Success rate results across 20 tasks on AgileX Cobot Magic V2.0.

Method	AGX-Open DrawerButton	AGX-Move ButtonDrawer	AGX-Stack Boxes	AGX-Find TapeBox	AGX-Sweep Rubbish	AGX-Arrange Valves	AGX-Hang Scissors	AGX-Place Button	AGX-Close Toolbox	AGX-Gather Screws	-
UniVLA	25	15	0	0	0	0	25	25	20	0	-
RDT	70	75	20	60	0	30	0	60	0	0	-
GR00T-N1.5	85	75	20	75	0	45	0	80	0	0	-
π_0	85	85	0	60	0	45	0	0	0	0	-
$\pi_{0.5}$	35	70	40	80	15	35	0	70	35	30	-
XR-1 (ours)	90	80	45	75	25	45	80	85	90	30	-
	AGX-Find Circuit	AGX-Place BiscuitBox	AGX-Collect BasketTea	AGX-Place Screwdriver	AGX-Pour GearOil	AGX-Stack BrakePads	AGX-Mesh StackCup	AGX-Pour Wine	AGX-Hang WipeRag	AGX-Stack Bowls	Avg.
UniVLA	0	0	0	0	10	10	25	0	0	20	8.8
RDT	0	0	0	0	0	55	0	0	65	85	28.5
GR00T-N1.5	0	50	45	0	0	0	0	0	20	70	24.0
π_0	0	40	45	55	40	0	55	0	50	90	32.5
$\pi_{0.5}$	20	60	25	30	55	40	15	20	60	90	41.3
XR-1 (ours)	15	60	35	35	75	85	90	20	55	85	60.0

Results on AgileX Cobot Magic V2.0. Table 11 reports success rates on 20 tasks with the AgileX Cobot Magic V2.0. XR-1 achieves an average of 60.0%, nearly doubling π_0 (32.5%) and far surpassing UniVLA (8.8%). On challenging tasks such as *AGX-StackBrakePads* and *AGX-CloseToolbox*, it reaches 85–90%, while other methods collapse to near 0%. We attribute these gains to UVMC-driven representations, which provide richer supervision and stabilize multi-task optimization.

Table 12. Success rate results across 20 tasks on Single-Arm UR-5e.

Method	SUR-Find Tape	SUR-Move MilkCup	SUR-Stack Bowls	SUR-Open Drawer	SUR-Close Drawer	SUR-Insert ToyBlock	SUR-Place Chopstick	SUR-Stack Cubes	SUR-Stack Cup	SUR-Stack Plates	-
UniVLA	35	25	50	20	35	0	0	0	0	0	-
RDT	80	35	20	35	45	0	0	0	0	0	-
GR00T-N1.5	85	70	80	25	70	0	0	0	25	0	-
π_0	25	55	90	50	85	0	0	80	0	55	-
$\pi_{0.5}$	55	30	95	85	95	0	35	90	90	85	-
XR-1 (ours)	95	85	95	90	90	15	20	90	85	85	-
	SUR-Slide Drawer	SUR-Open UpperDrawer	SUR-Open Oven	SUR-Pack EggBox	SUR-Close Laptop	SUR-Insert Bread	SUR-Assemble Valve	SUR-Pour TubeBeaker	SUR-Pour GearOil	SUR-Wipe HangRag	Avg.
UniVLA	30	30	35	0	35	0	30	0	20	30	18.8
RDT	40	35	55	15	50	0	15	10	35	30	25.0
GR00T-N1.5	45	65	80	0	90	0	80	0	10	30	37.8
π_0	75	90	55	20	85	30	20	10	45	75	47.3
$\pi_{0.5}$	90	90	95	80	90	85	25	10	50	75	67.5
XR-1 (ours)	80	90	90	70	90	65	90	20	85	75	75.3

Results on Single-Arm UR-5e. Table 12 summarizes success rates over 20 tasks on the Single-Arm UR-5e. XR-1 achieves the highest average success of 75.3%, clearly surpassing π_0 (47.3%) and all other baselines. XR-1 maintains strong performance (65% and 85%) on hard tasks like *SUR-InsertBread* and *SUR-StackPlates* where baselines often collapse to near 0%. These results highlight the robustness and generalization ability of XR-1 enabled by UVMC.

Table 13. Success rate results across 20 tasks Dual-Arm UR-5e.

Method	DUR-Find TapeBasket	DUR-Move CupMilk	DUR-Stack Bowls	DUR-Sweep Trash	DUR-Trans CupHolder	DUR-Stack Cubes	DUR-Hang CupHolder	DUR-Stack Brake	DUR-Sweep Rubbish	DUR-Press Button	-
UniVLA	30	35	35	20	0	0	0	0	0	10	-
RDT	45	65	50	0	30	0	60	15	10	20	-
GR00T-N1.5	25	60	80	0	0	0	0	0	0	0	-
π_0	70	55	80	55	15	15	10	0	0	35	-
$\pi_{0.5}$	45	65	90	60	55	20	10	85	70	80	-
XR-1 (ours)	85	65	85	60	65	20	85	65	80	85	-
	DUR-Pick PlaceTape	DUR-Close Toolbox	DUR-Assemble Valve	DUR-Flip TennisTube	DUR-Place Tools	DUR-Close DoorKnob	DUR-Take BasketTea	DUR-Pick Toolbox	DUR-Open TrashBin	DUR-Trans Buttons	Avg.
UniVLA	35	30	25	20	30	40	25	30	30	0	19.8
RDT	85	10	0	35	0	20	85	0	0	0	26.5
GR00T-N1.5	55	0	50	45	65	80	90	55	85	0	34.5
π_0	20	85	0	70	35	85	80	70	55	20	42.8
$\pi_{0.5}$	65	90	45	70	65	50	90	75	75	35	62.0
XR-1 (ours)	85	90	55	80	65	90	90	75	85	30	72.0

Results on Dual-Arm UR-5e. In addition to the bar plot reported in Figure 4, we provide the corresponding numerical results in Table 13. The table summarizes success rates across 20 tasks on the Dual-Arm UR-5e, offering a more detailed comparison among different methods.

E. Additional Ablation Study

Ablation study of UVMC. To obtain deeper insights into the UVMC architecture and its key hyperparameter choices, we conduct 10 ablation experiments, as summarized in Table 14, with Exp.10 serving as the baseline. By comparing Exp.1–6 against Exp.10, we analyze the influence of different codebook category numbers and code dimensions on the final performance. For the code dimension, Exp.1–3 adopt 64, 128, and 512, respectively, and are compared with the baseline setting of 256 in Exp.10. The results show that when the dimension is below 256, policy performance consistently improves as the dimension increases from 64 to 128 and 256, while further increasing it to 512 yields no clear additional gains, suggesting that a dimension of 256 is already near-optimal. Using a similar protocol in Exp.4–6 for the category number, we finally adopt 256 categories and a 256-dimensional codebook as our default configuration. Next, by comparing Exp.7–8 with Exp.10, we evaluate the difference between using only motion codes, only vision codes, and the unified vision–motion code (UVMC). The results indicate that both motion-only and vision-only variants underperform the unified UVMC. Moreover, vision-only codes outperform motion-only codes, while combining both modalities within UVMC leads to complementary effects and improved overall performance. Finally, by comparing Exp.9 and Exp.10, we investigate whether a combined or separate codebook is more effective. The results show that both designs achieve comparable performance, which we attribute to the alignment loss imposed during training: although a separate codebook increases the number of learnable codes, the alignment constraint effectively regulates cross-modal relationships, leading to similar execution capabilities for both schemes.

Table 14. Ablation study of UVMC.

Exp.	Codebook	Category×Embed.Dim	UVMC Token	Stage-1&2&3	DUR-Clean Table	DUR-Find TapeBasket	DUR-Move CupMilk	DUR-Stack Bowls	DUR-Sweep Trash	DUR-Trans CupHolder	Avg
1	combine	256×64	both	DT	35	55	50	60	35	45	46.7
2	combine	256×128	both	DT	45	65	60	70	55	65	60.0
3	combine	256×512	both	DT	55	75	50	85	65	55	64.2
4	combine	64×256	both	DT	45	60	55	65	35	50	51.7
5	combine	128×256	both	DT	50	65	55	65	60	60	59.2
6	combine	512×256	both	DT	40	80	60	80	55	65	63.3
7	combine	256×256	motion-only	DT	25	70	35	60	5	15	35.0
8	combine	256×256	vision-only	DT	10	70	65	70	15	65	50.0
9	separate	256×256	both	DT	55	75	55	85	40	80	65.0
10	combine	256×256	both	DT	50	75	65	80	60	70	66.7

Ablation study of Ego4d. To further examine the contribution of human video data (Ego4D) in the pre-training stage, we conduct a set of ablation experiments, as summarized in Table 15. To balance computational cost and the reliability of the conclusions, we use 10% of the full pre-training dataset for these comparisons. Under this setting, we evaluate two variants: one with Ego4D included in the pre-training data and one without Ego4D (w/o Ego4D). As shown in Table 15,

removing Ego4D leads to a 5.8% drop in average success rate compared to the setting that includes Ego4D. These results quantitatively suggest that incorporating Ego4D into the pre-training data can effectively improve performance.

Table 15. Ablation study of Ego4d.

Exp.	Instantiation	Stage-1	Stage-2	Stage-3	DUR-Clean Table	DUR-Find TapeBasket	DUR-Move CupMilk	DUR-Stack Bowls	DUR-Sweep Trash	DUR-Trans CupHolder	Avg
1	XR-1 w/o Ego4D	10%	XR-D	✓	20	60	10	55	15	35	32.5
2	XR-1 w/ Ego4D	10%	XR-D	✓	25	60	25	60	20	40	38.3

Cross-Embodied Knowledge Transfer for Enhanced Single Embodiment Performance. This setup is designed to verify whether similar tasks across different embodiments can mutually benefit each other. Since the UVMC counterpart of XR-1 learns an embodiment-agnostic feature, this setup serves to validate that capability. Specifically, we selected two identical tasks (FindTape and SweepRubbish) across three different embodiments (Dual-Arm Franka, Dual-Arm UR5e, and Tien Kung 2.0). The detailed results are shown in Table 16. Exp. 2 represents the results of training these two skills across three different embodiments, resulting in six tasks. In the comparative experiment setup, training two skills for a specific embodiment typically results in only two tasks. Therefore, to ensure fairness, in Exp. 1, we added four additional tasks for the same embodiment, ensuring that the data volume is equivalent. *The final results indicate that learning the same skills across different embodiments can enhance the success rate of each embodiment’s skills, increasing the average success rate by approximately 15%.* This demonstrates that the UVMC module has learned an embodiment-agnostic beneficial feature.

Table 16. Ablation study of XR-1 on cross-embodiment knowledge transfer.

Exp.	Instantiation	Stage-1	Stage-2	Stage-3	DFR-Find TapeBox	DFR-Sweep Rubbish	DUR-Pick PlaceTape	DUR-Sweep Rubbish	TK2-Take Tape	TK2-Sweep Rubbish	Avg.
1	XR-1	100%	XR-D	SelfRobot	50	20	70	50	60	30	47
2	XR-1	100%	XR-D	CrossRobot	70	30	70	60	70	70	62

F. Additional Generalization Analysis

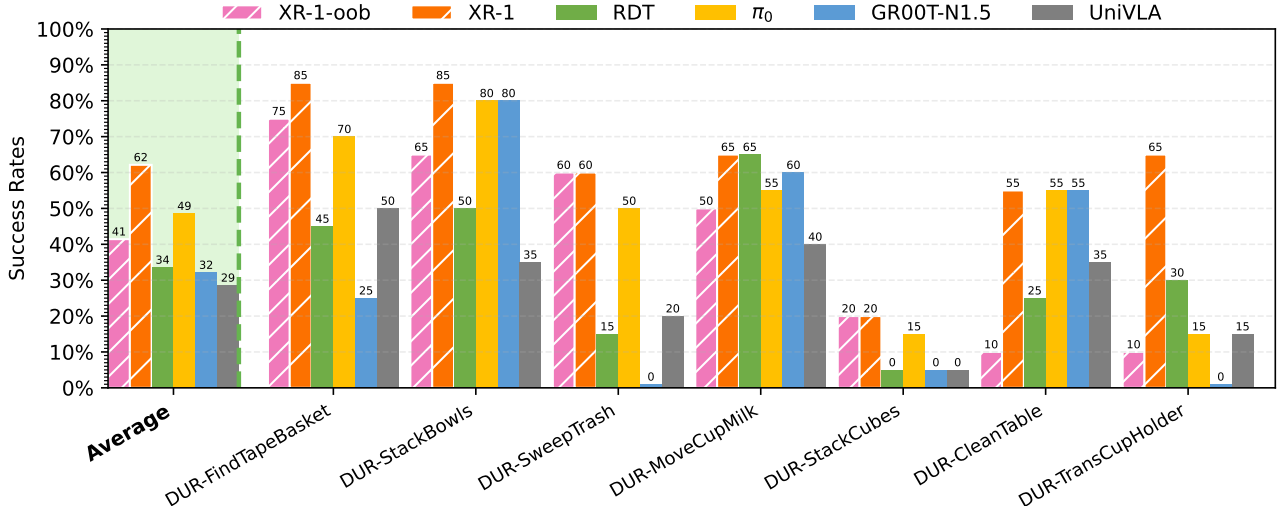


Figure 9. Out-of-box evaluation results of 7 tasks on Dual-Arm UR-5e.

Out-of-Box Evaluation. In addition to the evaluation on the Dual-Arm Franka, we also conduct an out-of-box evaluation of XR-1 on the Dual-Arm UR-5e. Specifically, we select 7 representative tasks from XR-D, covering only 0.45% of the dataset. To ensure a fair comparison, baselines without XR-D pretraining are fine-tuned on data from these tasks prior to evaluation. As shown in Figure 9, the pretrained **XR-1-oob** model, even without Stage-3 task-specific adaptation, achieves performance comparable to π_0 , while consistently outperforming GR00T-N1.5, RDT, and UniVLA. This result highlights XR-1’s strong generalization ability in low-data regimes.

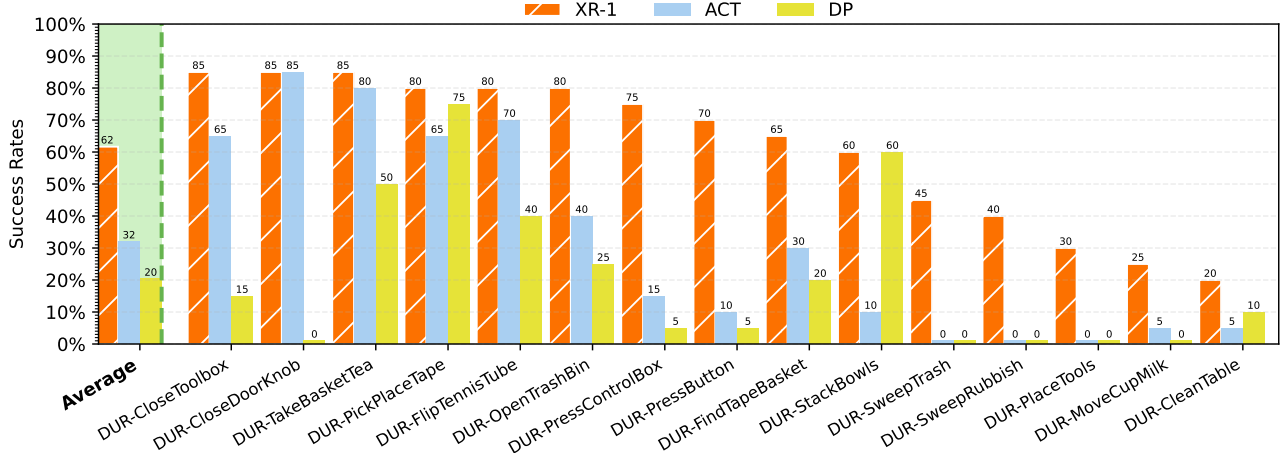


Figure 10. Fast adaption on Dual-Arm UR5e. Dual-Arm UR5e is an embodiment included in XR-D. In this setup, Here, XR-1 adapts to 15 novel tasks with one model using only 20-shot demonstrations per task, while baselines (ACT and DP) are trained per task.

Fast Adaptation to New Tasks. Beyond the experiments on Tien Kung 2.0, we also evaluate fast adaptation on the Dual-Arm UR-5e. Specifically, we collect 15 new tasks that are unseen in XR-D, each with 20 trajectories for training. XR-1 is trained jointly across these tasks, while single-task baselines, ACT (Zhao et al., 2023) and Diffusion Policy (DP) (Chi et al., 2023), are trained independently per task. As shown in Figure 10, XR-1 achieves substantially higher success rates than ACT and DP, even though the evaluation setting is more favorable to the baselines. This performance gain can be attributed to large-scale pretraining combined with UVMC supervision, which enables XR-1 to extract transferable representations from few-shot data and adapt effectively across diverse manipulation tasks.

G. Simulation Results on CALVIN

Simulation Benchmark. CALVIN (Mees et al., 2022) is a comprehensive simulated benchmark for evaluating language-conditioned policies in long-horizon robotic manipulation. It consists of four distinct yet closely related environments, denoted A, B, C, and D, which differ in desk textures and object layouts, as shown in Figure 11. The benchmark contains 34 manipulation tasks with unconstrained language instructions. Each environment features a Franka Emika Panda robot equipped with a parallel-jaw gripper, along with a tabletop scene containing a sliding door, a drawer, colored blocks, an LED, and a light bulb, all of which can be interacted with or manipulated.

Results. We report results on the challenging CALVIN benchmark under the D→D setting. This setting is particularly sensitive to error accumulation and the resulting distribution drift. As shown in Table 17, XR-1 achieves a new state-of-the-art success score of 4.256, outperforming strong baselines such as $\pi_{0.5}$ (3.885) and Qwen-GR00T (3.786). Notably, XR-1 maintains a success rate of 0.741 on the fifth sub-task, demonstrating stronger long-horizon robustness and improved resistance to distribution drift.

H. Visualization of Unified Vision-Motion Codes

H.1. Frame-to-Frame Nearest-Neighbor Retrieval between Motion and Vision Codes

To further evaluate whether the vision code (VC) and motion code (MC) are semantically aligned in the latent space, we design a nearest-neighbor retrieval experiment, as shown in Figure 12. We select 7 tasks, each with 10 trajectory episodes, and compute the VC and MC for all frames in each episode. At the current time step T , we extract the MCs corresponding to the actions GRASP and MOVE, as illustrated in the first column of Figure 12, and visualize the image of the current frame to better interpret the motion code semantics. We then compute VC features from different tasks and embodiments and, using cosine similarity, identify for each MC feature at time T its nearest-neighbor and farthest-neighbor VC feature. Columns two through five of Figure 12 report the cosine distances between MC and VC features under four settings: Same Embodiment with a Similar Task (SE.ST.), Same Embodiment with a Different Task (SE.DT.), Different Embodiment with a Similar Task (DE.ST.), and Different Embodiment with a Different Task (DE.DT.). For the different-embodiment setting, we use a dual-arm Franka robot. To standardize the notion of left and right across embodiments, we define them with respect to

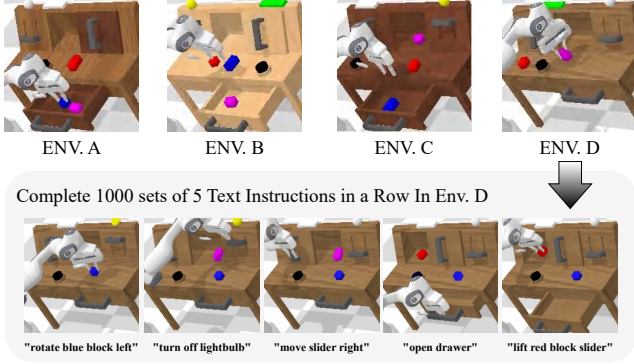


Figure 11. The upper four environments correspond to the CALVIN ABCD settings. The bottom section shows a sequence of five long-horizon tasks, each guided by a specific instruction.

Table 17. Comparison of XR-1 with baselines on CALVIN D→D.

Method	1	2	3	4	5	Success Rate
OpenVLA	0.716	0.385	0.180	0.088	0.041	1.411
RDT-1B	0.757	0.495	0.359	0.243	0.184	2.038
π_0	0.848	0.704	0.559	0.466	0.377	2.954
Qwen- π_0	0.909	0.795	0.696	0.622	0.554	3.576
Qwen-GR00T	0.925	0.839	0.744	0.679	0.599	3.786
$\pi_{0.5}$	0.925	0.840	0.766	0.710	0.644	3.885
XR-1 (ours)	0.964	0.908	0.845	0.798	0.741	4.256

the outward-facing direction of the embodiment.

Comparing SE.ST. with GRASP and MOVE shows that, under the same embodiment and a similar task, the nearest-neighbor VC for a given MC consistently reflects the same action semantics, whereas the farthest-neighbor images display clearly different motions. Under the SE.DT. setting, despite task changes, the nearest-neighbor VCs still capture the semantics of grasp or move, while the farthest-neighbor images correspond to distinct motions. In the DE.ST. setting, even with different embodiments, the nearest-neighbor frames for a similar task consistently depict similar actions, in contrast to the clearly different motions in the farthest-neighbor images. Likewise, in the DE.DT. setting, nearest-neighbor retrieval continues to select frames whose semantics are closest to grasp or move, despite differences in both embodiment and task, whereas the farthest-neighbor images represent dissimilar actions. Together, these visualizations demonstrate that UVMC successfully learns semantic representations of different actions, and that these representations become embodiment-agnostic.

H.2. Task-to-Task Nearest-Neighbor Retrieval between Motion and Vision Codes

In the preceding analysis (Appendix H.1), we primarily evaluate nearest-neighbor retrieval performance at the frame-to-frame level for representative skills (e.g., grasp, move) under different experimental settings (SE.ST., SE.DT., DE.ST., DE.DT.). To further investigate cross-task similarity from an episode-to-episode perspective, we construct the nearest-neighbor similarity distribution as illustrated in Figure 13. Specifically, we select four distinct tasks and perform pairwise comparisons among them, following the same experimental configurations as before (SE.ST., SE.DT., DE.ST., DE.DT.). Specifically, let the motion code of the i -th frame in the source task be denoted by MC_i , and the vision code of the j -th frame in the target task be denoted by VC_j . We first compute the similarity between each MC_i in the source task and all VC_j in the target task:

$$\mathbf{S} \in \mathbb{R}^{T_s \times T_t}, \quad \mathbf{S}_{i,j} = \text{sim}(MC_i, VC_j),$$

where T_s and T_t denote the number of frames in the source and target tasks, respectively. Based on this matrix $\mathbf{S}$, we perform a column-wise maximization, i.e., for each target frame j , we select from all source frames the motion-vision pair that attains the highest similarity:

$$s_j = \max_{1 \leq i \leq T_s} \mathbf{S}_{i,j}.$$

In this way, we reduce the original two-dimensional similarity matrix to a similarity vector:

$$\mathbf{s} = [s_1, s_2, \dots, s_{T_t}],$$

which characterizes, for each target frame, its nearest-neighbor similarity with the source task. Finally, we normalize the elements of $\mathbf{s}$ to obtain a normalized nearest-neighbor similarity vector $\tilde{\mathbf{s}}$, whose values are constrained to lie within $[0, 1]$, enabling comparable and stable statistical analysis across different tasks.

In Table 18, we report pairwise comparison results across different tasks. We take task 1 as the reference and use cosine similarity to quantify representational similarity between tasks. Under the SE.ST. setting, the mode of the similarity distribution is close to 1, indicating that for identical embodiments and tasks, the model learns highly consistent representations between MC and VC. Among the remaining three settings, DE.ST. has the highest mode, suggesting that UVMC learns features that

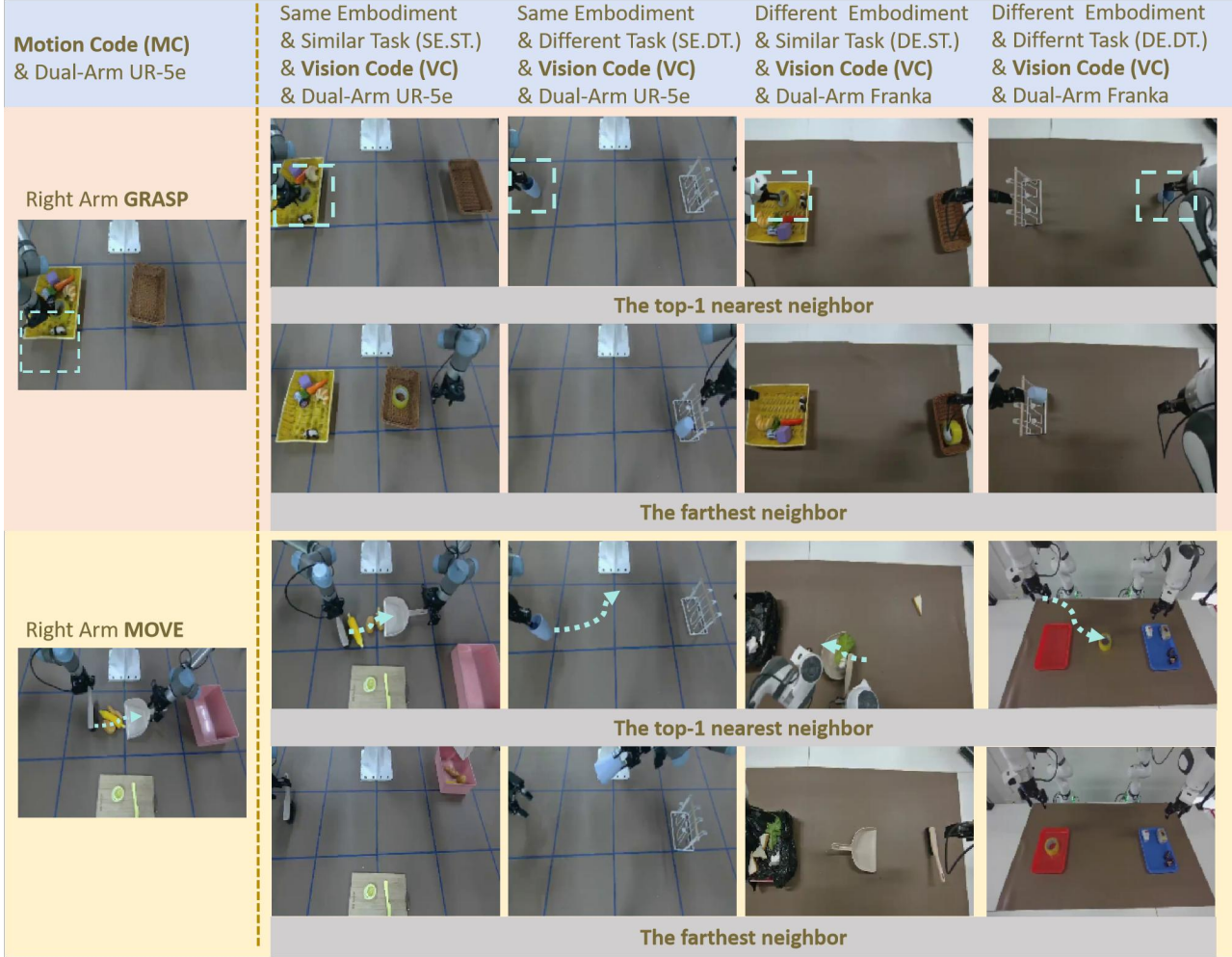


Figure 12. Frame-to-Frame Nearest-neighbor Retrieval between motion and vision codes. In the first column, we select two representative skill actions, grasp and move, and compute their Motion Codes (MC). Columns two through five report the cosine distances between Motion Code (MC) and Vision Code (VC) features and their nearest and farthest neighbors under four settings: Same Embodiment with a Similar Task (SE.ST.), Same Embodiment with a Different Task (SE.DT.), Different Embodiment with a Similar Task (DE.ST.), and Different Embodiment with a Different Task (DE.DT.). For the different-embodiment setting, we use a dual-arm Franka robot. To standardize the notion of left and right across embodiments, we define them with respect to the outward-facing direction of the embodiment.

are largely independent of embodiment and instead capture action-centric skills. Comparing DE.ST. with SE.DT. further supports this: different-embodiment but similar-task pairs exhibit higher overall similarity than same-embodiment but different-task pairs, implying stronger semantic alignment for shared skills than for shared morphology alone. Although DE.DT. has the lowest mode, it still retains non-trivial similarity, indicating that shared low-level skills (such as pick, handover, and place) give rise to stable cross-task, cross-embodiment similarity in the learned representations. We also compute the average nearest-neighbor similarity under these settings and observe consistent conclusions. Consequently, these analyses demonstrate that UVMC learns semantically meaningful action representations that are largely invariant to embodiment.

H.3. Visualizing UVMC with t-SNE

To qualitatively validate whether UVMC effectively captures intrinsic task dynamics and abstracts away physical embodiment details, we employ t-SNE to project the high-dimensional latent embeddings into a two-dimensional manifold. We conduct this analysis on two distinct subsets: (1) a single-robot scenario involving 6 tasks performed by a dual-arm UR robot (Figure 14), and (2) a mixed-embodiment scenario comprising both dual-arm Franka and dual-arm UR robots (Figure 15).

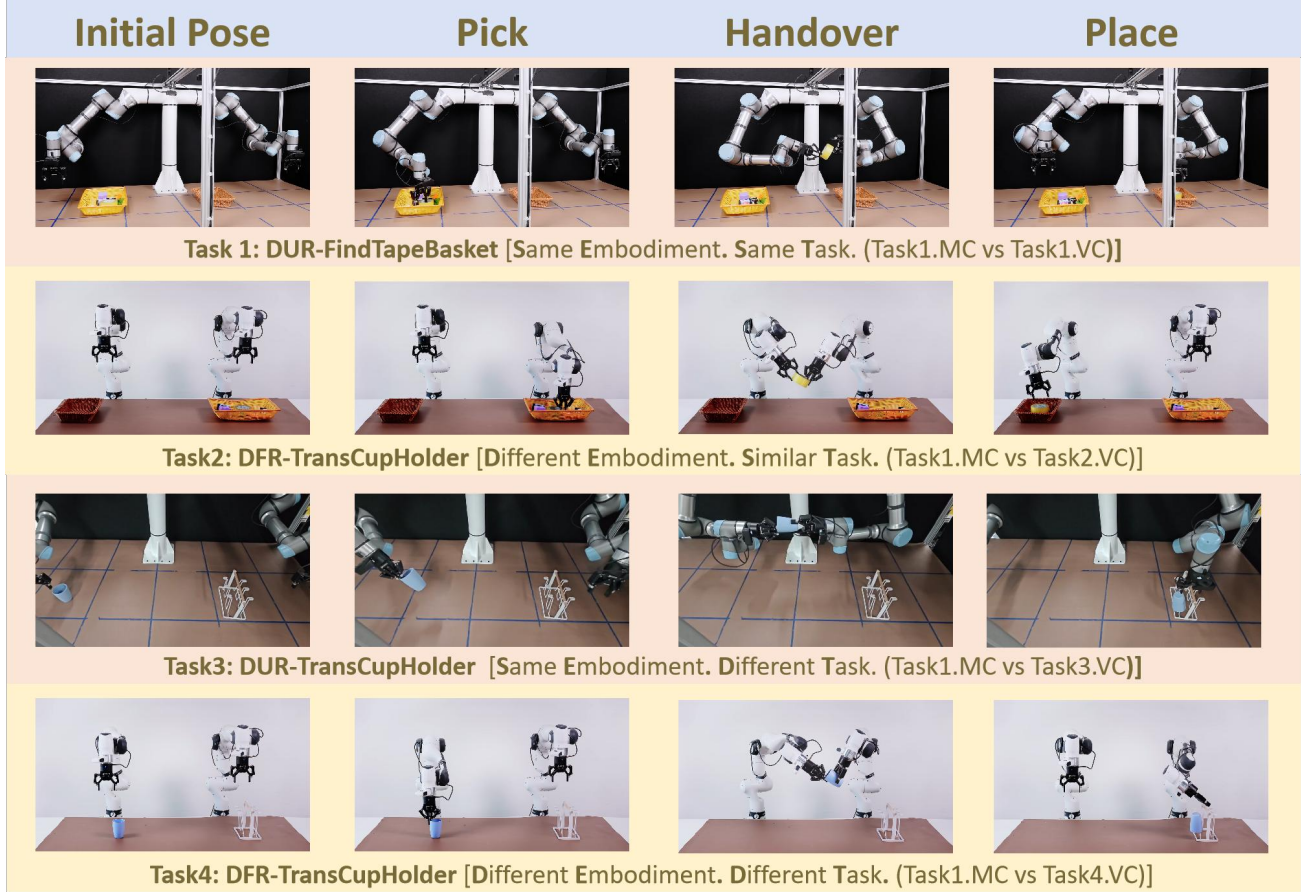


Figure 13. **Task-to-task nearest-neighbor retrieval between motion and vision codes.** It illustrates four different tasks and annotates the corresponding process phase at each timestep. For each task, the annotations correspond one-to-one to the four experimental settings in Table 18, representing the actual procedure used to compute nearest-neighbor retrieval across different tasks.

The visualization reveals two critical properties of the representation learned by UVMC.

Semantic Consistency of Dynamics. As shown in Figure 14, the embedding space exhibits a structured organization where tasks characterized by similar motion primitives form cohesive, proximal clusters. Specifically, in the lower-right corner, tasks sharing the "left-arm pick" primitive—namely DUR-CleanTable, DUR-StackBowls, and DUR-SweepTrash—are grouped closely together. This suggests the model successfully encodes the shared underlying semantics of the grasping motion. Conversely, the model effectively isolates distinct behaviors. Observing the upper region of the plot, a cluster of brown points forms a distinct island separate from other task embeddings. This cluster corresponds to the sweeping and translational motions unique to the DUR-SweepTrash task. This separation demonstrates that UVMC can effectively disentangle common dynamical patterns (e.g., picking) from task-specific nuances (e.g., sweeping).

Cross-Embodiment Alignment. A central hypothesis of our approach is that the learned representation should be embodiment-agnostic. Figure 15 substantiates this by illustrating the embedding space for identical tasks executed by morphologically distinct robots. Notably, the embeddings for DFR-CleanTable (represented in red) and DUR-CleanTable (represented in yellow) share a common support and overlap significantly within the latent manifold. Despite the kinematic and appearance discrepancies between the Franka and UR robots, UVMC projects their state-action trajectories onto a unified manifold. This result indicates that the model has learned a robust, embodiment-invariant representation that prioritizes high-level task semantics over low-level proprioceptive differences.

Table 18. Task-to-task Nearest-neighbor similarity statistics

Exp.	The mode of similarity distribution	Average
SE.ST. (Task1.MC vs Task1.VC)	0.97	0.75
DE.ST. (Task1.MC vs Task2.VC)	0.82	0.67
SE.DT. (Task1.MC vs Task3.VC)	0.60	0.56
DE.DT. (Task1.MC vs Task4.VC)	0.46	0.48

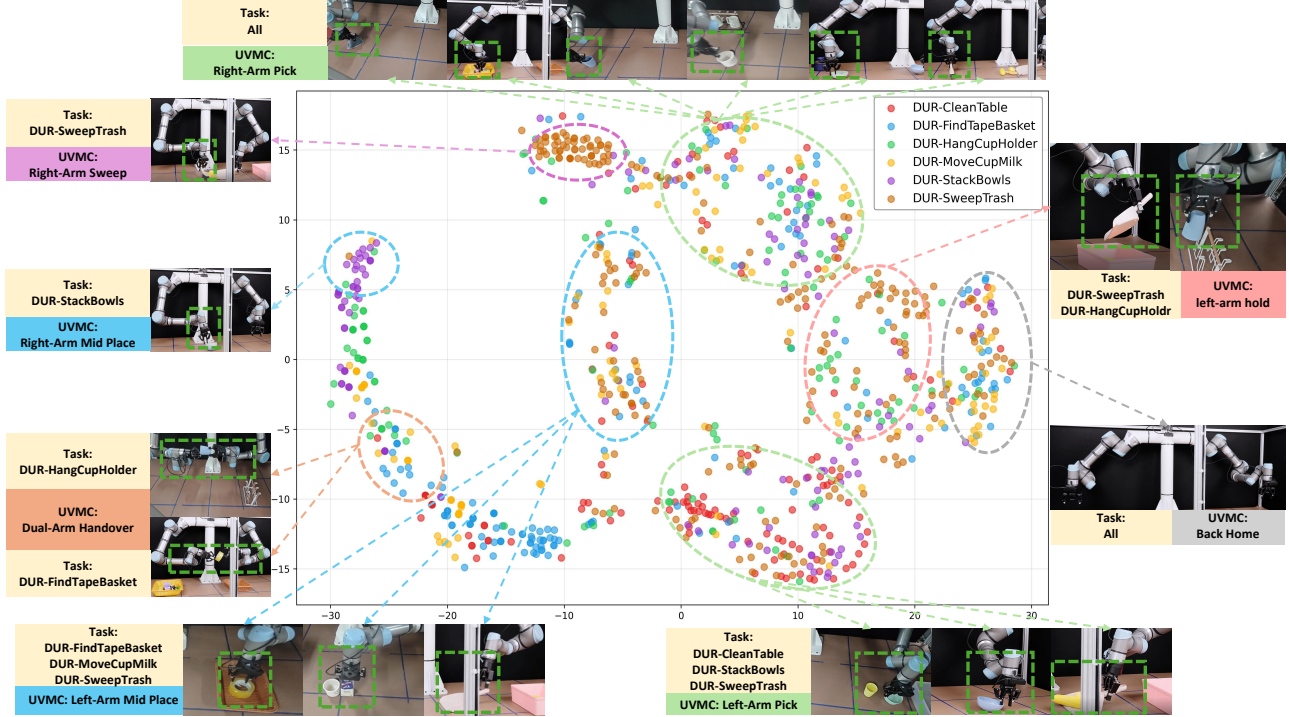


Figure 14. Visualizing UVMC on 6 dual-arm UR tasks using t-SNE.

I. Failure Case Analysis on Baseline Methods

We conducted a qualitative analysis of the rollout videos to investigate why baselines struggle compared to XR-1. We categorize the observed failures into two primary modes:

Optimization Collapse and Conflicting Gradients. In some distinct scenarios, baselines fail to capture the correct motion trend entirely. The models exhibit “hesitation” or revert to the mean pose, suggesting that the optimization objective is torn between conflicting gradients from different tasks. Example: In the ‘DFR-CloseToolbox’ task, the π_0 policy initiates a downward movement with the right arm but immediately retracts to the initial position. The robot appears indecisive and fails to commit to the task trajectory. We attribute this to the difficulty of fitting a single policy distribution to 20 diverse tasks without task-distinguishing representations.

Precision Deficiency and Coordination Failure. The most common failure mode involves the robot attempting the correct action but failing in execution precision or bimanual coordination. Example: In the ‘DUR-HangCupHolder’ task, GR00T-N1.5 successfully grasps the cup with the right arm. However, it drops the cup during the handover to the left arm. This indicates that while the model learns the general policy distribution, it lacks the fine-grained control and temporal consistency required for complex, multi-stage manipulation.

XR-1 addresses these issues through the Unified Vision-Motion Condition (UVMC) introduced in Stage 1. The UVMC serves as a compact representation of visual dynamics and motion patterns. By conditioning the policy on UVMC, XR-1 can explicitly distinguish between different task modes, thereby reducing gradient conflicts during multi-task optimization. As

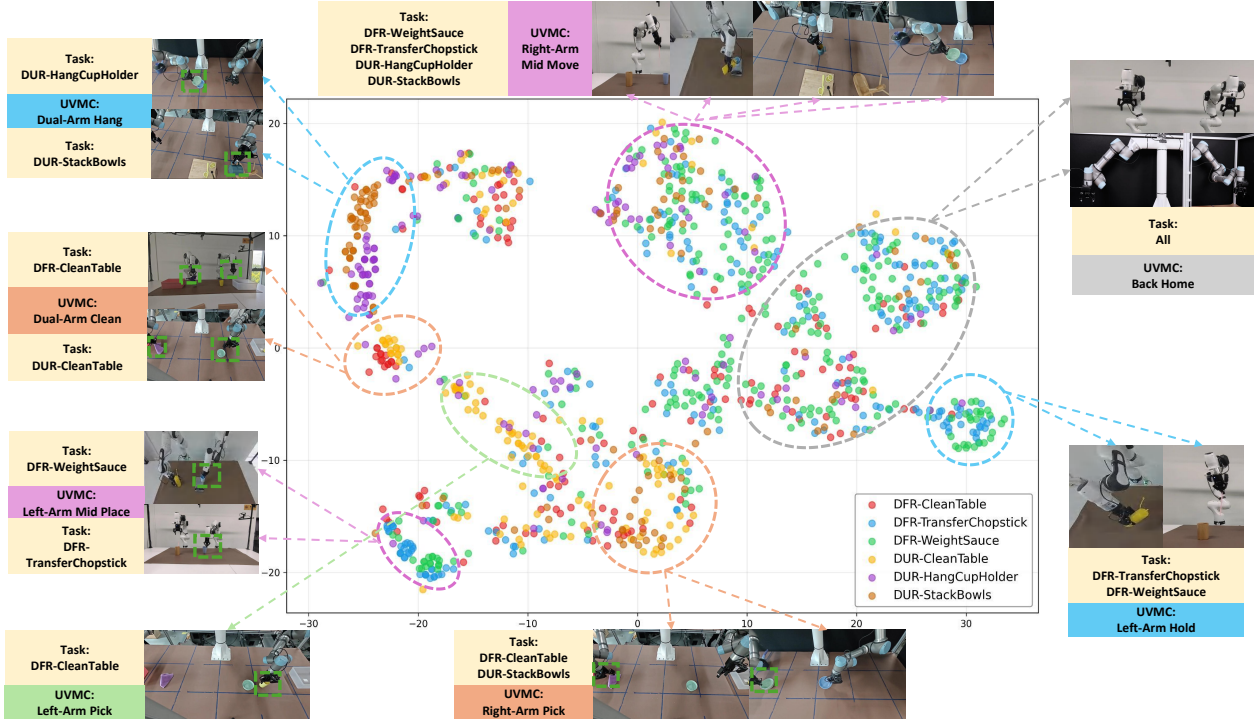


Figure 15. Visualizing UVMC across different embodiments (Dual-Arm Franka and Dual-Arm UR) using t-SNE.

an intermediate feature supervision signal, UVMC guides the model to generate smoother and more physically consistent actions. This additional supervision is critical for tasks requiring high precision (e.g., dual-arm handover), preventing the coordination failures observed in baselines like GR00T.

J. Failure Case Analysis on XR-1

Precision Deficiency. The most common failure mode involves the robot attempting the correct action but failing in execution precision or bimanual coordination. Example: In the ‘TK2-CollectScrews’ task, the robot may fail to grasp a tiny screw or drop it mid-motion due to slight localization errors. This reflects the inherent difficulty of learning precise bimanual coordination for dexterous manipulation tasks.

K. Representative Tasks

As illustrated in Figure 18, we select a set of representative tasks from real-world experiments to provide detailed descriptions of the evaluation scenarios. These tasks are designed to cover a broad spectrum of challenges, including bimanual collaboration, dexterous manipulation, fluid/deformable object handling, contact-rich interactions, dynamic environments, and long-horizon manipulation. Together, they demonstrate the versatility and robustness of **XR-1** across diverse manipulation settings.

- **Bimanual Collaboration:** *DUR-TransCupHolder*. This task involves a coordinated bimanual operation: the right arm initially grasps a cup, performs an aerial handover to the left arm, which subsequently places the cup into a cup rack.
- **Dexterous Manipulation:** *DUR-CloseDoorKnob*. The robot performs a dexterous operation to close and lock the control box door. The right arm first manipulates the door to a closed position. Subsequently, the left arm rotates the door handle by 90 degrees and presses it inward to engage the locking mechanism.
- **Fluid Object Handling:** *SUR-PourTubeBeaker*. The task consists of three phases: removing a test tube from the rack, pouring its liquid into a measuring cup, and returning the test tube to the rack.

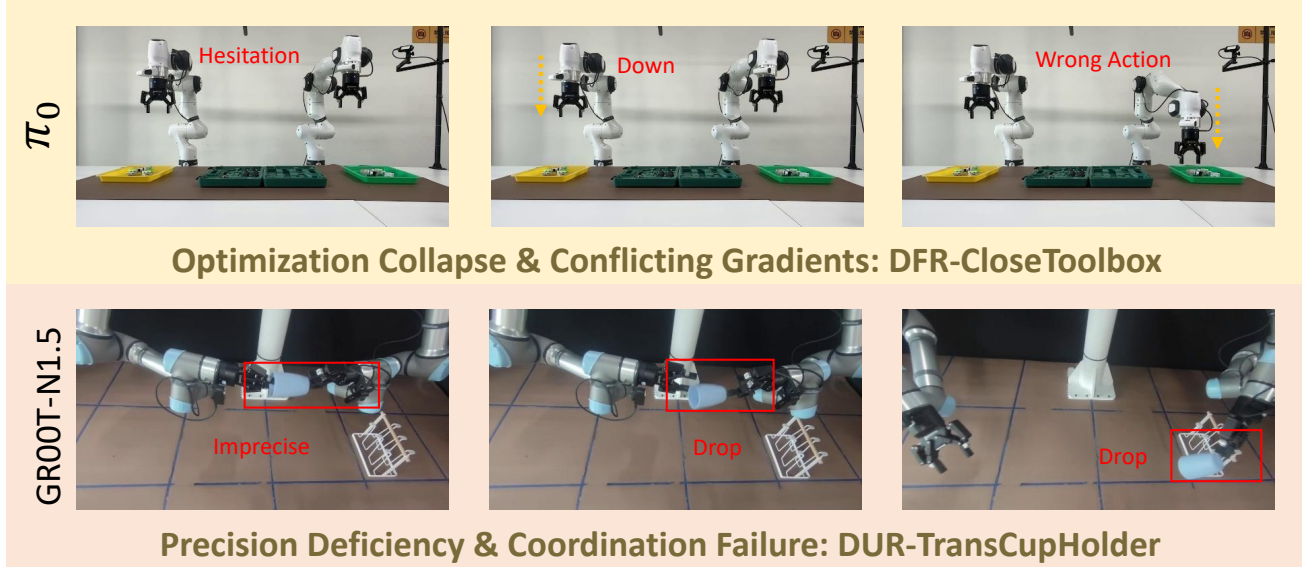


Figure 16. Failure cases of baseline methods.



Figure 17. Failure Cases of XR-1.

- **Deformable Object Handling:** *DFR-HangTowelRack*. The robot performs a bimanual manipulation task involving deformable object handling: the right arm first picks up a towel from a surface and transfers it to the left arm via an aerial handover; the left arm then manipulates the towel to drape it over a towel rack, completing the hanging motion.
- **Contact-Rich Interactions:** *DFR-SweepRubbish*. A dual-arm cleaning task is executed where the right arm operates a broom and the left arm stabilizes a dustpan. The robot systematically sweeps food remnants and a crumpled paper ball into the dustpan, followed by transporting and emptying the dustpan into a waste bin after each collection.
- **Dynamic Environments:** *DUR-TransButtons*. The robot’s left arm loads colored button workpieces onto a moving conveyor belt, while the right arm autonomously identifies each part’s color upon arrival and places it into the respective color-matched container.
- **Long-Horizon Manipulation:** *AGX-StoreButton*. This task entails a sequential dual-arm interaction: the left arm opens a drawer and holds it open, enabling the right arm to place a button workpiece inside; the left arm then closes the drawer after object deposition.

K.1. Dataset for Evaluation

The dataset is primarily employed for the final fine-tuning stage of XR-1, and for training and evaluation of multiple baselines on this benchmark.

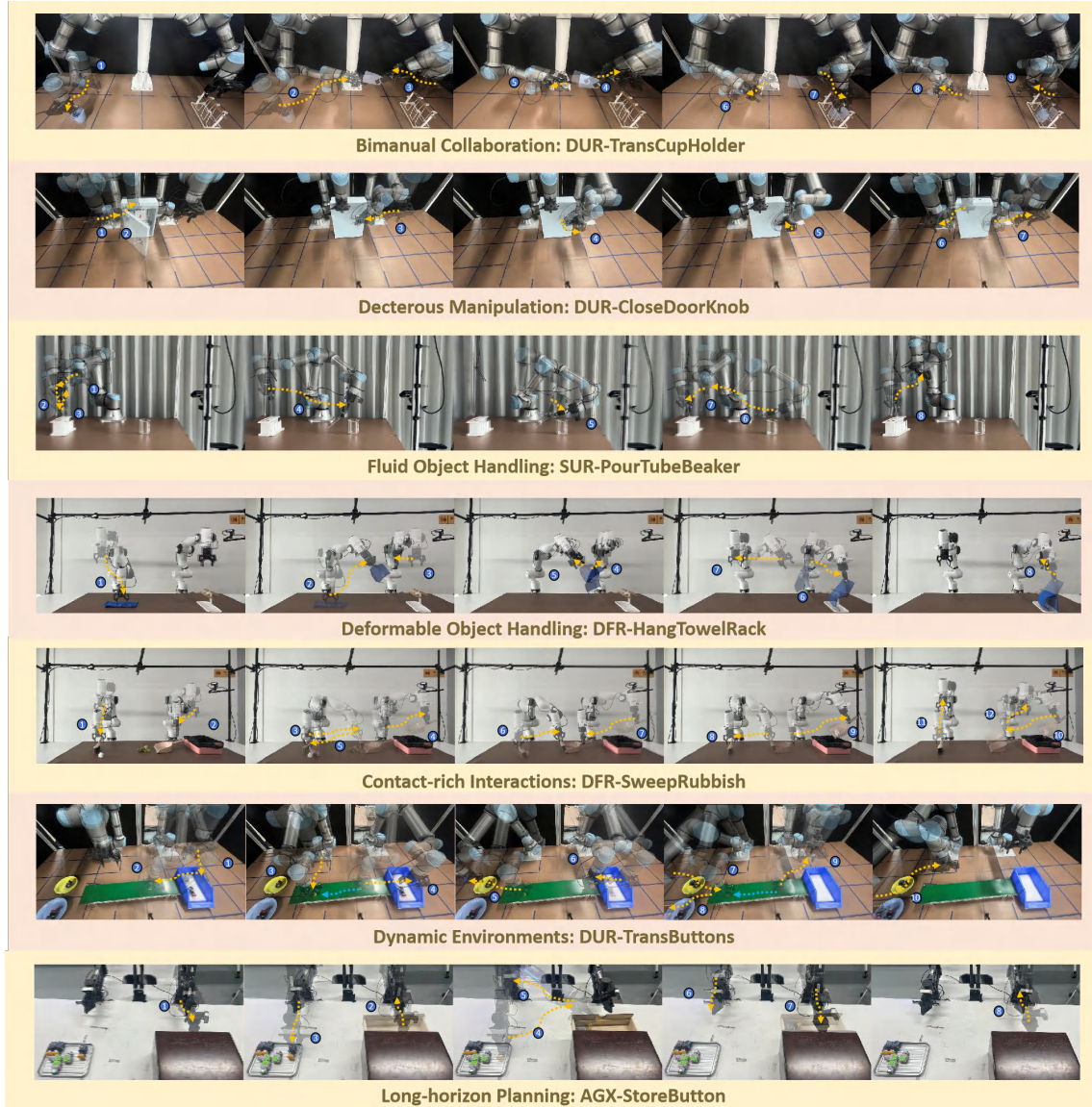
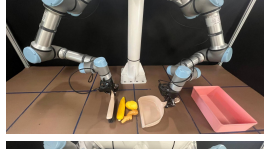
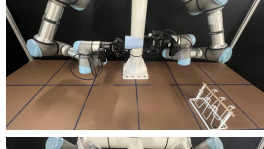
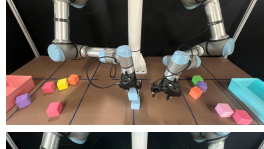
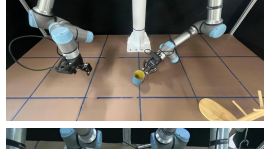
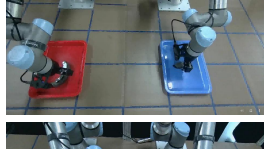


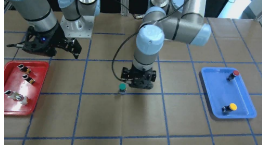
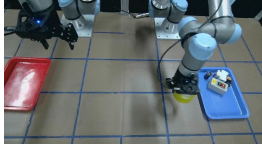
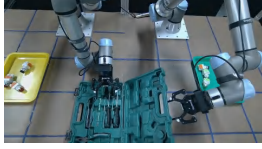
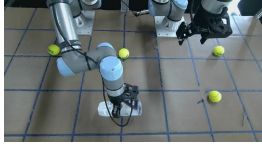
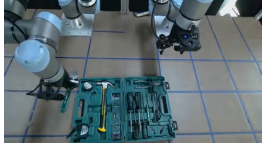
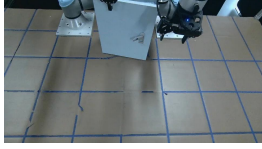
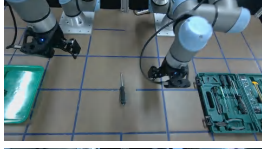
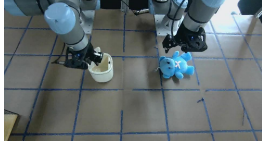
Figure 18. Diverse task settings in evaluation: bimanual collaboration, dexterous manipulation, deformable object handling, contact-rich interactions, dynamic environments, and long-horizon tasks.

Table 19. The tasks summary of our real-world experiments.

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
Dual-Arm UR5e					
T1	DUR-FindTapeBasket	160	155	Find the packaging tape and put it into the other basket	
T2	DUR-MoveCupMilk	198	149	Place the cup in the middle of the table and pick up the milk and place it next to the cup.	
T3	DUR-StackBowls	158	147	Put the blue bowl in the middle of the table and stack the green bowl on top of it	
T4	DUR-SweepTrash	192	293	Sweep up the rubbish and take out the trash	
T5	DUR-TransCupHolder	167	170	Pick up the cup with the right arm, hand it over to the left arm, and hang it on the holder with the left arm	
T6	DUR-StackCubes	158	153	Put the blue cube in the middle of the desk and stack it on top of the other blue cube	
T7	DUR-HangCupHolder	167	223	Hang the cup on the holder	
T8	DUR-StackBrake	200	81	Use the left arm to place Brake Pad Type A in the middle, then use the right arm to pick up Brake Pad Type B and stack it on top of Brake Pad Type A	
T9	DUR-SweepRubbish	185	157	Sweep up the rubbish	

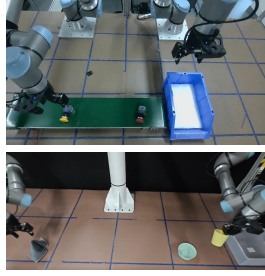
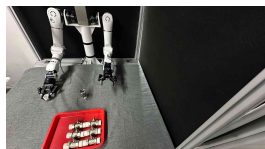
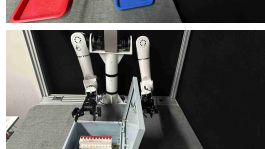
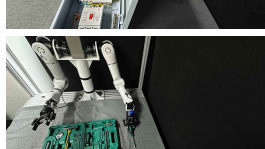
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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T10	DUR-PressButton	183	121	Pick up and place the green button, then press it	
T11	DUR-PickPlaceTape	192	111	Pick up and place the adhesive tape	
T12	DUR-CloseToolbox	114	190	Use both arms to close the toolbox	
T13	DUR-AssembleValve	253	108	Assemble the valve	
T14	DUR-FlipTennisTube	123	127	Put the tennis tube upright	
T15	DUR-PlaceTools	198	102	Use the left arm to place the screwdriver on the left side of the toolbox, and use the right arm to place the wrench on the right side of the toolbox	
T16	DUR-CloseDoorKnob	201	147	The right arm closes the distribution box door, and the left arm turns and presses the closing knob	
T17	DUR-TakeBasketTea	198	147	The right arm places the shopping basket in the middle, while the left arm takes tea drinks from the shelf and puts them inside	
T18	DUR-PickToolbox	200	117	Use both arms to pick up the toolbox	
T19	DUR-OpenTrashBin	28	45	Open the trash bin	

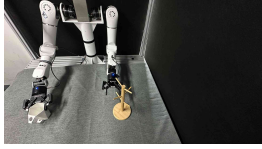
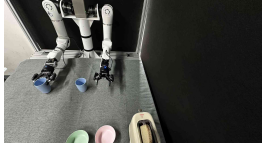
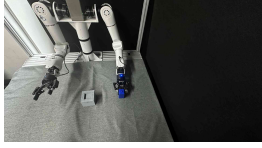
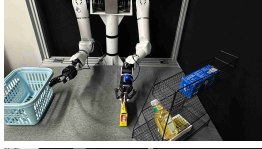
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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T20	DUR-TransButtons	146	184	Transport the buttons from left to right and place them in the corresponding plates	
T21	DUR-CleanTable	197	183	Move the buttons from left to right and place them on the corresponding plates	
Tien Kung 2.0					
T1	TK2-PressButton	291	292	The left arm picks up the green button and places it in the middle, while the right arm presses it	
T2	TK2-AssembleValve	162	382	Assemble the valve	
T3	TK2-StackBrake	178	261	Use the left arm to place Brake Pad Type A in the middle, and use the right arm to pick up Brake Pad Type B and stack it on top of Brake Pad Type A	
T4	TK2-PlaceCircuit	209	359	The left arm picks up the circuit breaker from the red tray and places it in the middle of the table. Then the right arm picks up the circuit breaker and puts it into the blue tray on the right.	
T5	TK2-PressControlBox	256	292	The left arm places the control box in the middle, while the right arm presses the red emergency stop button on it	
T6	TK2-CloseDoorKnob	197	271	The right arm closes the door of the distribution box, and the left arm rotates and presses the closing knob.	
T7	TK2-GatherTools	177	452	Use the left arm to place the screwdriver on the left side of the toolbox, and use the right arm to place the wrench on the right side	
T8	TK2-CloseLaptop	189	186	Close the laptop	

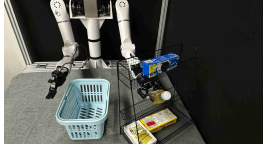
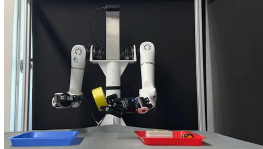
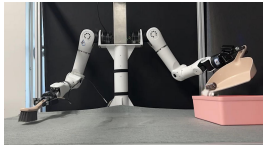
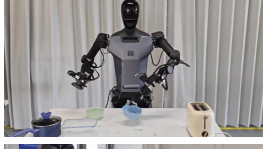
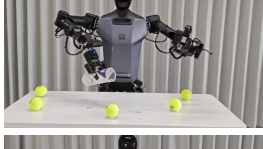
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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T9	TK2-OpenPotLid	190	304	Open the blue pot lid	
T10	TK2-HangCupHolder	144	277	Hang the oval-bottom cup on the holder	
T11	TK2-MoveCupSauce	129	312	Move the blue cup, pick up the yellow sauce bottle, and pour it into the blue cup	
T12	TK2-StackCup	161	291	Stack the blue cups	
T13	TK2-InsertToyBlock	152	521	Insert the blue toy into the square-bottom slot of the grey block	
T14	TK2-FindCapacitor	232	370	The left arm picks up the red electrolytic capacitor from the blue tray and places it in the middle of the table. Then the right arm picks it up and puts it into the red tray on the right	
T15	TK2-MoveMilkMug	279	284	Pick up and place the milk, then move the white mug	
T16	TK2-PourGearOil	152	503	The left arm places the gear on the middle metal tray, while the right arm pours lubricating oil on it	
T17	TK2-PlaceBiscuitBox	133	447	Pick up the biscuit box from the blue basket with the right arm and place it in the middle of the table. Then, use the left arm to place it on the middle shelf of the black rack	
T18	TK2-CollectScrews	192	620	The right arm places the two long screws into the slot at the very right end of the storage box, while the left arm places the two short screws into the slot at the very left end of the storage box	

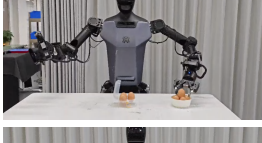
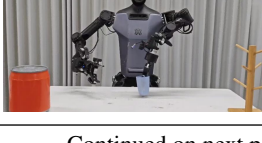
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#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T19	TK2-MoveTape	182	547	Pick up and place the rattan basket	
T20	TK2-TakeBasketTea	197	497	The right arm places the shopping basket in the middle, while the left arm takes tea drinks from the shelf and puts them inside	
T21	TK2-TakeTape	297	357	Pick up and place the adhesive tape	
T22	TK2-SweepRubbish	239	421	Sweep up the rubbish	
Tien Kung 1.0					
T1	TK1-FindTape	446	793	Find the packaging tape, pick it up, and place it into another basket	
T2	TK1-StackBowls	260	497	Put the blue bowl in the middle of the table, then stack the green bowl on top of it	
T3	TK1-CloseDrawer	231	223	Slide the drawer closed	
T4	TK1-FlipTennisTube	300	535	Put the tennis tube upright	
T5	TK1-PressCookerButton	98	187	Press the rice cooker's off button	
T6	TK1-MoveChopstickCup	274	574	Move the blue cup to the middle, then place one chopstick from the bamboo holder into it	

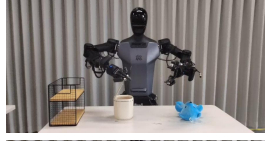
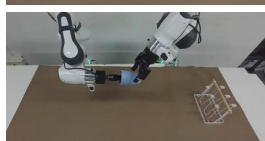
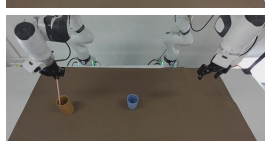
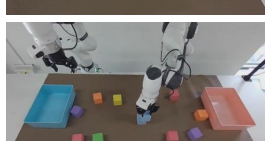
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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T7	TK1-StackCubes	200	520	Stack the two blue cubes	
T8	TK1-StackCups	200	487	Move the blue cup and stack it with the other blue cup	
T9	TK1-StackPlates	200	449	Place the pink plate into the beige plate in the middle, then stack the blue plate on top of the pink plate	
T10	TK1-PickWipeTowel	222	643	Pick up a towel and wipe the water with it	
T11	TK1-HangTowel	242	620	Pick up the towel with the right arm, hand it over to the left arm, and hang it on the rack with the left arm	
T12	TK1-OpenPotLid	104	486	Open the pot lid	
T13	TK1-OpenOven	21	228	Open the oven	
T14	TK1-PackEggBox	192	315	Put the egg into the box and close the lid	
T15	TK1-CloseLaptop	174	188	Close the laptop screen	
T16	TK1-InserToaster	174	254	Insert the bread into the toaster	
T17	TK1-FlipCup	153	578	Flip the cup upright	

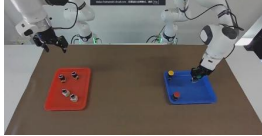
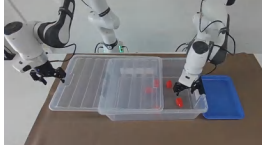
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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T18	TK1-PlaceFlipButton	125	616	Pick up the button with the right arm and place it in the middle. Then use the left arm to flip the button upright	
T19	TK1-OpenTrashBin	177	206	Open the trash bin	
T20	TK1-PressMachine	181	213	Press down the bread machine with the right arm	
Dual-Arm Franka					
T1	DFR-MoveCupMilk	293	254	Place the cup in the middle of the table, then pick up the milk and put it next to the cup	
T2	DFR-StackBowls	298	245	Put the blue bowl in the middle of the table and stack the green bowl on top of it	
T3	DFR-SweepTrash	248	350	Sweep up the rubbish and take out the trash	
T4	DFR-TransferCup	244	196	Pick up the cup with the right arm, hand it over to the left arm, and hang it on the holder with the left arm	
T5	DFR-MoveChopstick	277	299	The left arm moves the blue cup from the left side of the robot to the middle, while the right arm takes a chopstick from the bamboo holder on the right side and puts it into the blue cup	
T6	DFR-StackCubes	245	166	Put the blue cube in the middle of the desk and stack it on top of the other blue one	
T7	DFR-StackPlates	360	230	Use the left arm to place the pink plate into the beige plate in the middle, then use the right arm to stack the blue plate on top of the pink plate	

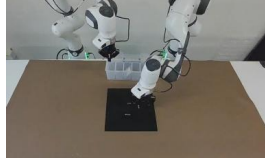
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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T8	DFR-CleanTable	284	201	Put the trash into the trash can, and put the items back in the box	
T9	DFR-HangCupHolder	206	199	Hang the cup on the cup holder	
T10	DFR-HangTowelRack	232	195	Pick up the towel with the right arm, hand it over to the left arm, and hang it on the rack with the left arm	
T11	DFR-FindTapeBox	194	245	Find the packaging tape and put it into the other box	
T12	DFR-PickButtonPress	200	206	The left arm picks up the green button and places it in the middle, while the right arm presses it	
T13	DFR-SweepRubbish	196	288	Sweep up the rubbish	
T14	DFR-CloseToolbox	201	220	Use both arms to close the toolbox	
T15	DFR-CollectBasketTea	200	186	The right arm places the shopping basket in the middle, while the left arm takes tea drinks from the shelf and puts them inside	
T16	DFR-PlaceTools	199	159	Use the right arm to place the wrench on the right side of the toolbox, and use the left arm to place the screwdriver on the left side	
T17	DFR-GetBlocks	94	255	The right arm grabs the storage box and opens the lid, while the left arm places the red building blocks inside, ensuring they do not fall off	

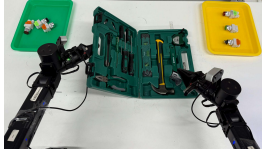
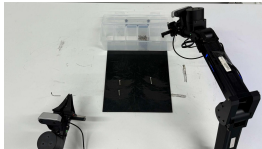
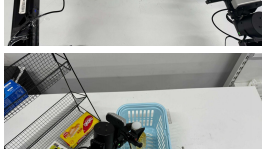
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#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T18	DFR-PlaceRagWipe	196	301	Use the right arm to place the rag in the middle of the table, and use the left arm to wipe the remaining liquid on the middle of the table with the rag	
T19	DFR-OpenToolbox	199	244	Use both arms to open the toolbox	
T20	DFR-PlaceScrews	189	301	The right arm places the two long screws into the slot at the very right end of the storage box, while the left arm places the two short screws into the slot at the very left end of the storage box	
AgileX Cobot Magic V2.0					
T1	AGX-OpenDrawerButton	272	1418	Slide open the drawer and place the yellow button inside	
T2	AGX-MoveButtonDrawer	387	1612	Place the yellow button in the drawer and close it	
T3	AGX-StackBoxes	169	1514	Put the left box in the middle, then stack the right box on top of it	
T4	AGX-FindTapeBox	182	1058	Find the packaging tape and put it into the other box	
T5	AGX-SweepRubbish	112	2370	Sweep up the rubbish	
T6	AGX-ArrangeValves	194	1411	Arrange the valves in a row	

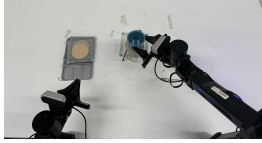
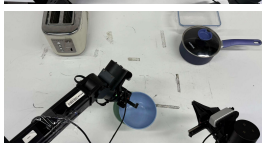
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#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T7	AGX-HangScissors	48	3124	Hang the scissors on the holder	
T8	AGX-PlaceButton	184	1794	Take the blue tray and place a button on it	
T9	AGX-CloseToolbox	184	2003	Use both arms to close the toolbox	
T10	AGX-GatherScrews	152	2918	The right arm places the two long screws into the slot at the very right end of the storage box, while the left arm places the two short screws into the slot at the very left end of the storage box	
T11	AGX-FindCircuit	476	2880	The left arm picks up the circuit breaker from the red tray and places it in the middle of the table. Then the right arm picks up the circuit breaker and puts it into the blue tray on the right	
T12	AGX-PlaceBiscuitBox	188	1983	Pick up the biscuit box from the blue basket with the right arm and place it in the middle of the table. Then, use the left arm to place it on the middle shelf of the black rack	
T13	AGX-CollectBasketTea	190	3078	The right arm places the shopping basket in the middle, while the left arm takes tea drinks from the shelf and puts them inside	
T14	AGX-PlaceScrewdriver	177	3079	The right arm picks up the Phillips screwdriver and places it in the middle of the table. Then, the left arm picks it up again and puts it into the groove in the toolbox	
T15	AGX-PourGearOil	192	3144	The left arm takes the gear and places it on the middle metal tray, and the right arm pours lubricating oil on the gear	

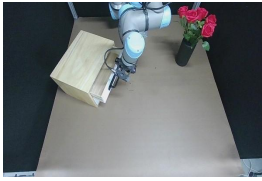
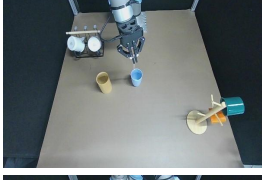
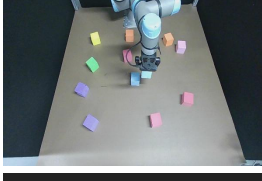
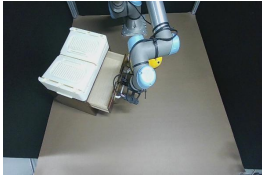
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#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T16	AGX-StackBrakePads	188	1650	Use the left arm to place Brake Pad Type A in the middle, and use the right arm to pick up Brake Pad Type B and stack it on top of Brake Pad Type A	
T17	AGX-MeshStackCup	117	1765	Place the mesh and stack the cup on it	
T18	AGX-PourWine	162	1644	Pour the wine with the right arm and place the cup on the tray with the left arm	
T19	AGX-HangWipeRag	199	2257	Use the right arm to place the rag in the middle of the table, and use the left arm to wipe the remaining liquid with it	
T20	AGX-StackBowls	185	1314	Stack the blue bowl on top of the green bowl	
Single-Arm UR5e					
T1	SUR-FindTape	134	104	Find the packaging tape and put it into the other basket	
T2	SUR-MoveMilkCup	292	116	Pick up the milk and place it next to the cup	
T3	SUR-StackBowls	300	118	Stack the blue bowl on top of the green bowl	
T4	SUR-OpenDrawer	212	168	Slide open the drawer	

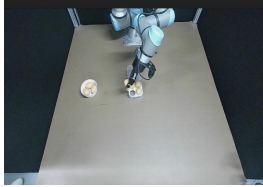
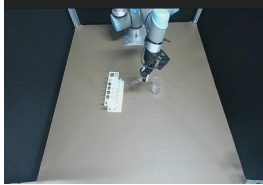
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#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T5	SUR-CloseDrawer	190	191	Slide the drawer closed	
T6	SUR-InsertToyBlock	150	198	Insert the blue toy into the square-bottom slot of the grey block	
T7	SUR-PlaceChopstick	297	109	Place one chopstick from the bamboo chopstick holder into the blue cup	
T8	SUR-StackCubes	308	92	Stack the two blue cubes on top of each other	
T9	SUR-StackCup	284	192	Stack the cups	
T10	SUR-StackPlates	291	194	Stack the plates in the middle	
T11	SUR-SlideDrawer	56	181	Slide open the drawer	
T12	SUR-OpenUpperDrawer	182	132	Open the upper drawer	

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#	Task	Trajectory		Task Instruction	Task Setting
		Num.	Avg. Len.		
T13	SUR-OpenOven	141	70	Open the oven	
T14	SUR-PackEggBox	183	151	Put the egg into the box and close the lid	
T15	SUR-CloseLaptop	196	107	Close the laptop screen	
T16	SUR-InsertBread	193	156	Insert the bread into the toaster	
T17	SUR-AssembleValve	182	166	Assemble the valve	
T18	SUR-PourTubeBeaker	152	270	Pick up the test tube and pour water into a 50 ml glass beaker	
T19	SUR-PourGearOil	172	214	Take the gear and place it on the middle metal tray, then pour lubricating oil on it	
T20	SUR-WipeHangRag	209	134	After using the rag to wipe the water in the middle of the table, hang the rag on the rag rack	